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Survey paper

Fruit and vegetable disease detection and classification: Recent trends, challenges, and future opportunities

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ABSTRACT

Fruits and vegetables are major sources of nutrients for the majority of the population across the globe. With the rapid increase in population, the objectives of the future agro-industry are to reduce product loss while increasing product quality and productivity considerably. Consequently, farmers need to be assisted with cutting-edge technologies for sustainable, eco-friendly, and efficient farming. Smart farming for early disease recognition and control is the current hot-spot research objective in the fruitage domain. The precision agriculture era has been revolutionized by federating cutting-edge technologies like machine learning, deep learning, and, the Internet-of-Things. However, the existing studies focused on the impact of individual technology on single or multiple cultivars of edible fruits or vegetables. Limited areas of the fruitage disease remain explored, necessitating further investigation into the research gaps and challenges identified for implementing the smart practices in real-field farmlands. In this paper, a comprehensive survey of recent advancements in fruit and vegetable disease identification and classification is presented. The technology-wise state-of-the-art findings, gaps, challenges, and future opportunities for fruitage disease recognition have been presented, covering 99 research articles. Moreover, the corpus of publicly available fruit and vegetable datasets has been investigated, with the existing gaps, improvements, and future requirements. The research paper concludes with challenges and a future outlook that promises to be a very significant and valuable resource for researchers working in the area of agronomic disease monitoring.

1. Introduction

Agriculture and horticulture are the largest economic contributors in India. Approximately 58% of the Indian population still depends on agriculture for their livelihood (IBEF, 2023). About 20.19% Gross Domestic Product (GDP) of the Indian economy is contributed by the agricultural sector (GDP, 2022). As per the reports of the Food and Agriculture Organization (FAO), India is the second largest global producer of fresh fruits and vegetables, yielding 12% of overall production (ICAR, 2022). National Horticulture Board Statistics (NHBS) recorded that India has produced a cumulative of 102.48 and 200.45 million metric tons of fruits and vegetables, respectively, in the year 2020–21 (APEDA, 2022). Fruit cultivation covers nearly 9.6 million hectares of land, while vegetable farming covers 10.86 million hectares of land. Fig. 1 shows the distribution of primary fruits and vegetables across different zonal regions of the country. Moreover, the FAO of the United Nations claimed that food security demands would be the world's prime concern in the upcoming years due to the rapid expansion of the world population. The Ministry of Food Processing Industries (MFPI) of India reported that approximately 33 million metric tons of fruit and vegetable costing 10.6 billion USD deteriorates annually due to lack of proper management and technology. The objective of the United Nations' sustainable development is to achieve "Zero Hunger by 2030" across the world. However, the COVID-19 pandemic made it harder to overcome the global nutrition crisis and showed the significance of technological support to confront hunger, malnutrition, and non-communicable diseases linked to diet (FAO, 2022b). India ranked 107 among 121 countries in the Global Hunger Index-2022 (GHI, 2022). The National Family Health Survey (2019–21) reported that around 200 million Indian children under five years were undernourished (Survey, 2022). Fruit and vegetable demand is jeopardized by the proliferation of disease, insects, pests, nutritional deficiencies, and fertilizer deficits, causing considerable production losses. The disease diagnosis at an early stage is the primary requirement for crop protection and epidemic control.

The conventional methods employ visual inspection and knowledgeable human experts for early disease detection, which are sometimes

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Fig. 1. Zonewise distribution of primary fruits and vegetables across different Indian States.

less available, especially for small-scale farmers in developing nations (Lee et al., 2020). Additionally, less experienced farmers are not able to make smart decisions regarding pest identification, resulting in a dwindling agriculture economy and efficacy. Moreover, ingesting microbially contaminated vegetables and fruits may pose a grave risk to human health. World Health Organization (WHO) reported that nearly 600 million individuals become ill annually due to food contamination (WHO, 2022). Consequently, this crucial problem necessitates incorporating innovative and practicable technical solutions to ensure the mechanism for the fruit disease detecting system. Additionally, it will be fruitful in packaging, quick transportation, valuation, dietary guidance, and consumer health. Agriculture 4.0 has revolutionized the agriculture industry with the alliance of several intelligent and innovative techniques, thus helping agrarians in improving the quantity, fertility, and quality of fruits and vegetable crops (Sharma et al., 2022). With the continuous development of precision farming techniques, agriculture, and other allied sectors have compensated for the production loss by implementing autonomous disease detection techniques using Computation Intelligence (CI), real-time sensors, computer vision, Unmanned Aerial Vehicles (UAV), and Neural Networks (NN) to overcome the issue of the fruit production scarcity, time lacking, high cost and requirement of extensive skilled and experienced labors. In a nutshell, precision farming plays a crucial role in acquiring timely and precious information about disease prediction, thus providing effective methods for aiding decision-making and intelligent management in several fruit and vegetable domains such as autonomous harvesting robots, grading and sorting, category identification, real-time fruit disease classification and prediction, and quality evaluation.

Recently, a number of machine learning, deep learning, and Internet-of-Things-based techniques have been introduced by several researchers for early prediction of fruit and vegetable disease. Intelligent technological support for early disease identification is the need of the hour for sustainable agricultural development. Moreover, finding the current research trends, future opportunities, and optimal models based on the parameters, datasets, experimental scenarios, and multiple hardware configurations is still a challenging task for the researchers (Dhaka et al., 2021). As an effective data-driven framework, machine learning models and other imaging techniques offer precise, generalized methods for complex relationships between massive ranges of variables (Sharma et al., 2020). Deep learning methods, especially Convolution Neural Networks (CNN) and their derivatives, have recently gained substantial attention and offered practical approaches to complex real-world agricultural problems. CNN trains using image features and automatically extracts contextual information and global descriptors, which significantly reduces errors and offers several benefits over conventional image processing techniques. Deep Transfer Learning (DTL) has also been significantly used to develop reliable models on customized datasets (Jin et al., 2020; Hasan et al., 2023). The sensor-generated data collected and communicated through the Internet-of-Things, and cyber-agriculture methods may be used by agrarians to monitor their fruit and vegetable crops remotely. As a result, the combination of contemporary UAV camera sensor technology and deep learning algorithms might be a viable approach for early fruit disease identification (Quy et al., 2022). Thus, an exhaustive literature survey is necessary to consolidate the recent technological advancements and challenges in smart agriculture for the fruit and vegetable industry. The overall contribution of the study is detailed as follows.

- A systematic survey covering the recent advancements in machine learning, deep learning, and Internet-of-Things for automatic fruit and vegetable disease detection using image processing and computer vision-based methods has been presented.
- The current research challenges, literature gaps, and future opportunities have also been investigated. Additionally, the pros, cons, results, and applications of the different state-of-the-art frameworks targeting the classification and identification of fruit and vegetable diseases using real-time and collected images.
- Moreover, the publicly available datasets for the fruit and vegetable disease classification have also been tabulated for the ease of the research community working in the field of smart horticulture.
- This study may serve as a powerful tool for researchers and agronomists in the development of innovative agriculture models for predicting fruit and vegetable diseases.

Table 1
Studies close to the current literature review.

References	Fruit species	Target application		Technology focused				Benchmark dataset
	Single (✓) Multi (✓✓)	Grading	Classification	ML	MCV	DL	IoT	
Zhang et al. (2018)	✓✓	✓			✓		✓	
Iqbal et al. (2018)	✓		✓	✓	✓			
Razmjooy et al. (2019)	✓	✓		✓	✓			
Chandrasekaran et al. (2019)	✓✓	✓			✓		✓	
Kalia et al. (2019)	✓✓	✓		✓				
Kantale and Thakare (2020)	✓		✓	✓	✓			
Alam Siddique et al. (2020)	✓	✓	✓	✓	✓			
Naranjo-Torres et al. (2020)	✓✓	✓	✓		✓	✓		
Nabi et al. (2020)	✓		✓		✓		✓	
Bhargava and Bansal (2020c)	✓	✓			✓		✓	
Behera et al. (2020)	✓✓	✓		✓	✓			
Antony and Kumar (2021)	✓✓	✓	✓	✓	✓			
Patil et al. (2021a)	✓	✓		✓	✓	✓	✓	
Saranya et al. (2021)	✓	✓	✓		✓	✓		
Patil et al. (2021b)	✓		✓	✓	✓			
Dhiman et al. (2022b)	✓✓	✓		✓	✓	✓		
Ukwuoma et al. (2022)	✓✓		✓		✓	✓		✓
Shruthi et al. (2022)	✓		✓	✓	✓	✓		
Singh et al. (2022)	✓✓	✓	✓	✓	✓	✓		
Sivaranjani et al. (2022)	✓✓	✓	✓	✓	✓			
Palei et al. (2023)	✓	✓	✓	✓	✓			
Rizzo et al. (2023)	✓✓	✓	✓	✓	✓	✓		
Current survey	✓✓	✓	✓	✓	✓	✓	✓	✓

ML = Machine Learning, **MCV** = Machine Computer Vision, **DL** = Deep Learning, **IoT** = Internet-of-Things.

The organization of the literature survey is structured as follows: Section 2 presents the research background followed by the introduction in Section 1, whereas selection & inclusion criteria of the research articles have been detailed in Section 3. Section 4 discusses past surveys on fruitage disease classification using machine learning. Section 5 details previous studies in the deep learning domain that have been utilized for fruit & vegetable disease classification & identification. Section 6 thoroughly reviews different fruitage disease detection & classification approaches using remote sensing technologies and Internet-of-Things applications. Section 7 defines the characteristics of different publicly available fruit & vegetable datasets from the previous research works. Section 8 discusses the open challenges and future opportunities in their respective fields. Section 9 summarizes the classification performance of the state-of-the-art methods. Finally, the paper is concluded in Section 10.

2. Research background and motivation

The economic growth rate of developing countries is closely tied to agriculture. Fruits and vegetables encompass a broad spectrum of plant foods that differ substantially in calorie and nutritional density. Fruits and vegetables are a great source of fiber, which has been linked to a reduced risk of cardiovascular disease and obesity. As per the reports from the FAO, about 30%–40% of the fruits and vegetables are lost due to pest attacks globally (FAO, 2022a). Moreover, the Food Corporation of India (FCI) has estimated a 10%–15% loss of overall production in the upcoming years. Therefore, accurate and timely estimation of disease severity, defects incidence, and negative consequences of disorders on the fruits and vegetables is crucial for improving pesticide efficacy, fruit & vegetable breeding, and productivity.

The traditional disease identification method includes studying visual pathogen symptoms (rots, blight, anthracnose) using fruit phenotype by trained subject experts, which is labor-intensive and tedious. Nevertheless, visible symptoms are subject to human bias and temporal variation, causing inaccurate and varied outcomes. Therefore, autonomous early disease recognition and detection have been identified as an emerging field in intensive plant research, which can aid in stopping disease spread in other parts of the fruits and vegetables like roots and leaves. To transform the conventional agriculture system to ecological agriculture 4.0, different disruptive intelligent technologies such as machine learning, deep learning, Internet-of-Things, drones, and wireless sensor networks are used to mitigate the risks associated with fruitage diseases (da Silveira et al., 2021) as picturized in Fig. 2. Ecological informatics bridges the gap between ecosystem and informatics by combining ecological values, leading-edge technologies, and massive environmental factors to better fruit and vegetable disease detection and pest control. Moreover, fruit pathogens thrive in an environment with ecological factors like high humidity and wetness, overwatering, and soil contamination, which causes the spread of bacteria, viruses, and fungus (Caleb et al., 2013). Fruit and vegetable diseases lower yield estimation and cause quality and quantity degradation, resulting in loss of commercial production. To remedy this, insecticides and pesticides are used, which may increase the toxic content in food products. Therefore, it is necessary to detect the disease early to minimize pesticide consumption. Table 1 presents literature studies performed in the last six years to automatically detect and classify fruit and vegetable disease, aggregating 22 available studies. Different researchers examine the impact of solely smart techniques for single and multiple cultivars of fruits and vegetables.

Iqbal et al. (2018) highlighted the impact of computer vision & machine learning algorithms for early disease detection in citrus species. Chandrasekaran et al. (2019) suggested a comprehensive survey about near-infrared and hyperspectral spectroscopy on multiple agro-products using computer vision imaging systems. Naranjo-Torres et al. (2020) urged the impact of CNN and image processing in the field of the fruit and vegetable domain. Bhargava and Bansal (2020c) introduced an exhaustive survey on the disease severity estimation of multiple apple cultivars using spectral imaging techniques. Behera et al. (2020) addressed the fruit grading and disease classification problem using computer vision and machine learning methods. Patil et al. (2021a) examined the grading and sorting of dragon fruit based on their maturity level in the computational intelligence field. Dhiman et al. (2022b) summarized the survey on the quality evaluation and future opportunities of the multiple cultivars based on the disease severity levels using machine and deep learning algorithms. Sivaranjani et al. (2022) presented an overview of the disease classification for varied agro-products. Rizzo et al. (2023) explored the different non-destructive methods for ripeness categorization in the fruit & vegetable industry. The problems of disease detection in fruit images persist due to considerable variations in natural surface color in different fruits and their

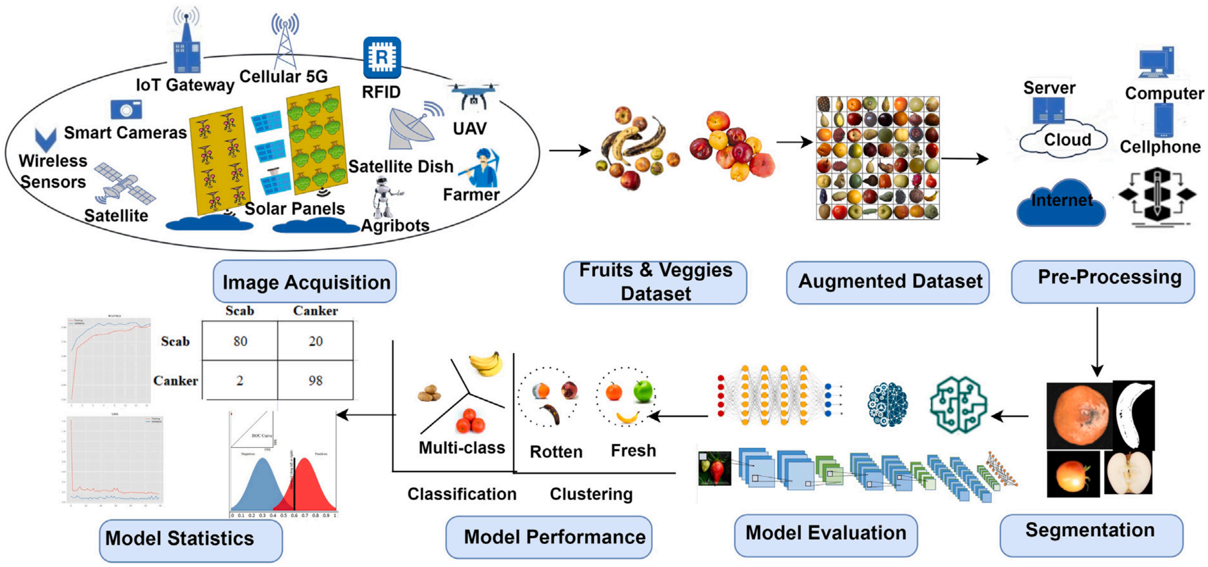

Fig. 2. Paradigms of different smart technologies utilized in precision agriculture.

species, color, and texture. However, no study in the literature has been performed, which includes an extensive survey covering state-of-the-art technologies for the automated disease detection of fruits and vegetables. Additionally, researchers find it hard to choose an effective model based on the parameters, datasets, experimental scenarios, and multiple hardware configurations due to the availability of a multitude of research works in implementing innovative agriculture models for predicting fruit and vegetable diseases (Dhaka et al., 2021). Thus, there is a need for an exhaustive literature survey from the existing research work that can help the researchers in identifying a suitable innovative model as per pre-processing techniques, forecasting, and classification of different diseases in multiple cultivars of fruits and vegetables. Thus, a Systematic Literature Survey (SLR) plays a tremendously significant role in understanding the controlling factors to avoid similar agro-product losses in the future. The primary contribution of the survey is to analyze the current fruitage industry scenario at the empirical and qualitative levels and disseminate information regarding the progress of Agriculture 4.0 at the global level.

3. Research methodology

The current review paper aims to investigate and present the survey of existing machine learning, deep learning, and Internet-of-Things-based methods for disease detection and monitoring in multiple cultivars of fruits & vegetables. The current research survey follows the standard methodology pattern to make the study work transparent and impartial regarding information selection and result representation (Keele et al., 2007). The study starts with research questions and their motivation and ends with the final selection of qualified research papers.

Study Selection Process: The selection of the research papers has been carried out using Preferred Reporting Items for Systematic reviews and Meta-analysis (PRISM) technique (Liberati et al., 2009) which contains (1) Initial identification; (2) Screening or introductory filtering; (3) Secondary filtering; and (4) Exclusion and inclusion. Table 2 addresses the research objectives and their respective motivation in the field of precision farming. It helps to design the present review paper's complete research framework, which can help to provide concise and relevant research information in the fruit & vegetable disease classification domain.

The next phase of the study is to search for significant existing research papers on the relevant topic for the custom year range from 2018 to 2023. A pilot search using specific keywords has been performed through multiple digital repositories and search engines like Web of Science and Google Scholar to collect related information about the fruitage disease framework. The repository list chosen for review work includes Elsevier, Springer, ACM, MDPI, IEEE Xplore, etc. The research papers were obtained using all possible combinations of auxiliary keywords that were screened to get the best source for the answers to the defined research objectives. Table 3 depicts the different search strings used to identify the pertinent research work.

After collecting the research papers, screening was performed to get the current trends in the fruit or vegetable disease recognition system as the standards suggested by Dybå and Dingsøyr (2008). The overall statistical flow of review papers selection is depicted in Fig. 3. The preliminary screening step includes the papers that match and satisfy the pre-defined research objectives and titles. The secondary screening is performed after reading the abstract of the papers collected during the primary phase. The abstract gives a brief idea about the proposed research methodology, datasets, and result comparison, thus, helping to make a notable repository. Initially, more than 200 papers were included from the aforementioned digital library based on the current technologies. The research papers having duplicate and irrelevant titles were eliminated to design an optimized library. The screening process has been employed to include the papers in the context of smart agriculture, satisfying research questions, search criteria, abstract, publishing year, and full article, resulting in 123 papers. For instance, most research papers were excluded due to plant leaf disease identification instead of edible fruit or vegetable itself. Therefore, the outcome of the study is 99 primary studies which will become the foundation of the review work.

Finally, the papers are segregated technology-wise. The following sections present the technology-wise findings of the research performed for fruit and vegetable disease identification.

4. Machine learning

Machine learning (ML) is a subdomain of computer intelligence that emphasizes useful knowledge-driven algorithms and self-learning of machines for good prediction and classification. These algorithms have the extrinsic capability for self-learning the various field applications affecting agriculture, such as disease detection, yield prediction, maturity level, evapotranspiration or water management, stress prediction, and fruit grading. Food defects are identified as abnormal changes that

Table 2
Research objectives.

Research objectives	Research questions	Motivation	Outcomes
RO1	What are the fundamental & reputed publication means for the current technologies in agro-products domain	It helps to identify the ethical publication sources for smart agriculture research work	Publication selection
RO2	What are the main focus areas in precision farming	It helps to explore the various keywords and application domains in agriculture like disease monitoring	Area & technology selection
RO3	How were the research work carried out filtered on year, country and author, journal	It helps to get knowledge about the current trends in the fruit disease recognition sector	Latest research categorization
RO4	What mathematical approaches and simulations are pre-existed to solve the fruitage disease classification issues	It helps to get insights into the different state-of-the-art architectures, reviews, platforms, and their implementations	Study of the selected papers
RO5	What are different datasets having multiple cultivars of fruits and vegetables available that are most frequently used	It focuses on data acquisition with multiple dimensional features	Study of publicly available datasets
RO6	What are the different performance criteria used in the evaluation of the intelligent models in the recent research	It helps to get insight knowledge about the comparative study of the past research	Result analysis
RO7	What are the potential conclusions and challenges of the relevant papers	It helps to study the opportunities, technical glitches, and their future gaps in the pre-qualified surveys	Epilogue and technical advancements

Table 3
Search strings.

Digital library	Search criteria			Sub domain	Domain
	Techniques	Objective	Year		
Elsevier, Springer, ACM, MDPI, IEEE Xplore, ProQuest, Google Scholar Web of Science, IGI Global, Mendeley, Kaggle, Github, Frontier in Plant Science, IEEE Data Port and ArXiv	Deep Learning, Machine Learning, Computer Vision, UAV, Cyber-Physical System and Internet-of-Things	(Disease OR Defects OR Infection OR Disorder OR Pathogen) AND (Classification OR Detection OR Identification OR Monitoring OR Severity)	2018–2023	Fruits and Vegetables	Smart Farming, Precision Agriculture, E-Agriculture, IoAT, Digital Farming, Agriculture 4.0

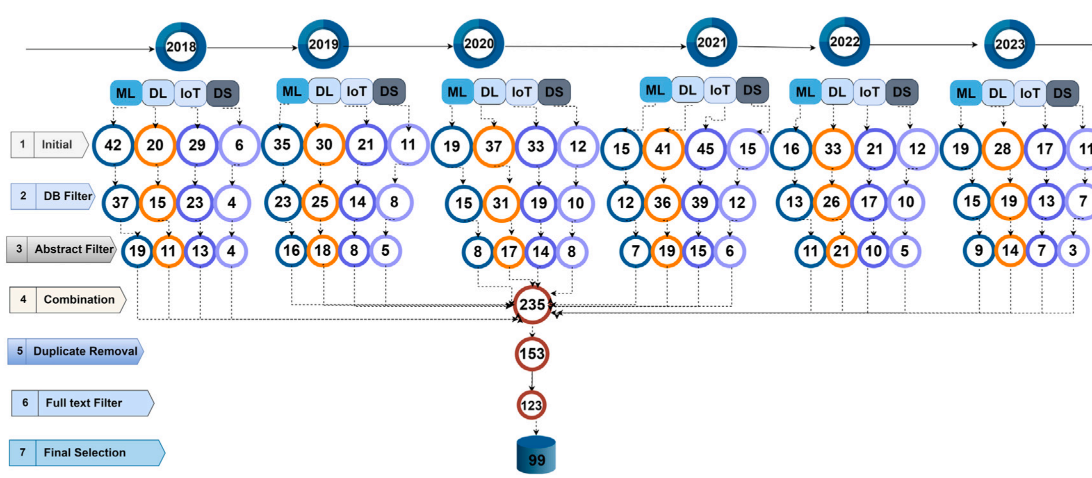

Fig. 3. Different steps involved in Systematic Literature Review.

may hamper the functional growth of the plant, which can be caused by some viruses, fungi, or bacteria. Early prediction of such pathological disorders could reduce food pre and post-harvest losses. However, farmers must ensure significant enhancement in food production by adopting leading-edge ML technologies in intelligent or precision farming. Decision tree (Dai et al., 2016), K-nearest neighbors (KNN) (Singh and Kaur, 2018), random forest (Chaudhary et al., 2016), and Support Vector Machine (SVM) (Yu et al., 2018), Principal Component Analysis (PCA) (Dhiman et al., 2021), Linear Discriminant Analysis (LDA) (Jahanbakhshi and Kheiralipour, 2020), Scale-Invariant Feature Transform (SIFT) (Langford et al., 2019), Histogram of Oriented Gradient (HOG) (Roy et al., 2021b), K-means (Bhargava and Bansal, 2020b), etc. are some of the classical exemplars of ML algorithm used for the fruit disease detection. Table 4 depicts the basic ML formulations used in the disease classification for the fruit & vegetable domains.

4.1. Fruit & vegetable disease detection using ML techniques

The manual fruitage disease inspection and grading are tedious as they suffer from poor accuracy due to unskilled human efforts. Recently, a number of ML-based methods have been introduced for the automated detection of fruit and vegetable diseases. Table 5 presents an

Table 4
Machine learning techniques.

Ml	Ref	Models	Description	Application	Variants	Formulation
Supervised learning	Quinlan (1996)	Decision Tree	It performs the attribute selection, determining the splitting and stopping criteria, carrying the pruning methods, & labeling terminal nodes.	Classification	ID3	$E = -\sum_{k=1}^{C} p_k \log_2 p_k$ $Gain - Ratio = E(X) - E(X/Y)$
					C 4.5	$E = -\sum_{k=1}^{C} p_k \log_2 p_k$ $Gain - Ratio = E(X) - E(X/Y)$
					CART	$Gini = 1 - \sum_{k=1}^{C} p_k^2$
					CHAID	$\chi^2 = \sum_{k=1}^{C} \frac{(O_k - E_k)^2}{E_k}$
					QUEST	$\chi^2 = \sum_{k=1}^{C} \frac{(O_k - E_k)^2}{E_k}$
	Goldstein et al. (2017)	Regression	It is a continuous-valued predictive model for finding the association between the dependent variable and set of independent feature vectors.	Regression	Linear	$y = \beta_1 x + \beta_0$
					Logistic	$\log_e \frac{y}{1-y} = \beta_1 x + \beta_0$
					Polynomial	$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$
					Lasso	$\sum_{m=1}^{C} (y_m - \sum_{n=1}^{C} x_{mn}\beta_n)^2 + \lambda \sum_{n=1}^{p} \lvert\beta_n\rvert$
					Ridge	$\sum_{m=1}^{C} (y_m - \sum_{n=1}^{C} x_{mn}\beta_n)^2 +$ $\lambda \sum_{n=1}^{p} \beta_n^2 continuous - valued$
	Zhu et al. (2017)	Ensemble	A generalized meta method for optimal performance by combining several standard models.	Classification	Random Forest	$RFfi_l = \frac{\sum_{j \in trees} normfi_{ij}}{T}$
					AdaBoost	$H(X) = sign(\sum_{i=1}^{T} \alpha_i * h_i(x))$
					XGBoost	$Obj(\theta) = \sum_{k=1}^{n} L(y_k, \hat{y_k}) + \Omega(f_t)$
	Bishop and Nasrabadi (2006)	Naive Bayes	It is Bayes theorem probabilistic classifier given with a "naive" hypothesis of conditional independence between each predictor feature.	Classification	Gaussian NB	$P(x_i \mid y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \frac{-(x_i - \mu_y)^2}{2\sigma^2}$
					Multinomial NB	$\theta = \frac{N_{yi} + \alpha}{N_y + \alpha * n}$
					Bernoulli NB	$P(x_i \mid y) = P(x_i = 1 \mid y)x_i + (1 - P(x_i = 1 \mid y))(1 - x_i)$
					Categorical NB	$P(x_i = t \mid y = c; \alpha) = \frac{N_{tic} + \alpha}{N_c + \alpha * n}$
	Garg and Mago (2021)	SVM	It optimizes the linearly separable hyperplane as decision boundary with maximum margin in n dimension.	Classification	Linear	$(weight)^T x + bias = 0$
					Multiquadric	$(\lVert x - \overline{x} \rVert^2 + \sigma^2)^{\frac{1}{2}}$
					Radial Basis	$exp(\frac{-\lVert x - \overline{x} \rVert^2}{2\sigma^2})$
					Polynomial	$(x^T x_i + 1)^p$
	Li and Wang (2014)	LDA	It maximizes the ratio of between & within class variance while retaining the class discrimination.	Dimension reduction	Fisher LDA	$J(\theta) = \frac{(\widetilde{\mu_2} - \widetilde{\mu_1})^2}{\widetilde{s_1}^2 + \widetilde{s_2}^2} = \frac{w^T S_B w}{w^T S_W w}$
					Null LDA	$W = argmax_{\lvert W^T S_W W = 0} \lvert W^T S_B W \rvert$
	Garg and Mago (2021)	KNN	It is instance-based learning where the maximal class likelihood is determined by predetermined 'k' nearest points.	Classification	Hamming	$H_d = \sum_{k=1}^{n} \lvert (x_i - y_i)^2 \rvert$
					Euclidean	$E_d = (\sum_{k=1}^{n} (x_i - y_i)^2)^{\frac{1}{2}}$
					Manhattan	$M_d = \sum_{k=1}^{n} \lvert (x_i - y_i)^2 \rvert$
Un supervised learning	Hotelling (1933)	PCA	It reduces the dimensions with a maximum variance while retaining the original data.	Dimension reduction	Linear	$J(e_1, e_2, \dots e_m) =$ $\sum_{a=1}^{m} e_a^t S e_a - \sum_{b=1}^{m} \lambda_j (e_b^t e_b - 1)$
					Gaussian Kernel	$\widetilde{K_g} = K - 2M_{\frac{1}{n}} K_g + M_{\frac{1}{n}} K_g M_{\frac{1}{n}}$
	Memarsadeghi and O'Leary (2003)	Clustering	It groups the whole data points into the number of clusters having identical data.	Segmentation	K-Means	$\sum_{k=1}^{N} \sum_{C(i)=k} \lVert x_i - \hat{\mu}_k \rVert^2$
					Fuzzy C-Means	$\sum_{i=1}^{C} \sum_{j=1}^{N} ij^m \lVert x_i - \hat{\mu}_k \rVert^2$
	Reynolds (2009)	GMM	It is parametric modeling for weighted addition of Gaussian probability densities.	Clustering	Maximum Likelihood	$p(x) = \sum_z p(z)p(x \mid z) = \sum_{i=1}^{n} \pi_i N(x \mid \mu_i, \Sigma_k)$
					Variational bayesian	$P(X \mid Z, \mu, \Sigma)P(Z \mid \pi)P(\pi)P(\mu \mid \Sigma)P(\Sigma)$
	Saha and Manickavasagan (2021)	ICA	It decomposes multivariate data to distinct & independent non-gaussian form	Blind source separation	Naive	$x = As = \sum_{i=1}^{n} a_i s_i$
	Alahmed et al. (2020)	SVD	It is a matrix factorization method decomposing any matrix into three matrices by Singular value	Dimension reduction	Naive	$M = U\Sigma(M^t M)^T$
Semi supervised learning	Van Engelen and Hoos (2020)	Inductive	It trains the model by labeling the unseen unlabeled data points. It generalize to unobserved samples	Classification	Self training	$(A_l, B_l) = (A_{1:l}, B_{1:l})$ and $A_u = (A_{l+1:n})$
					Co training	$L^i = L_i \bigcup max(x, p_j(x)) \forall i \neq j$
		Trans- ductive	It trains the model by labeling the already seen unlabeled data points. It does not generalize to unobserved samples	Classification	TSVM	$J(w, b) =$ $min[\frac{1}{2} \lvert\lvert w \rvert\rvert^2 + C \sum_{i=1}^{L} \epsilon_i + C^* \sum_{i=L+1}^{L+U} \epsilon_i]$
					Graph Learning	$w_{ij} = exp(\frac{\lVert x_i - x_j \rVert^2}{2\sigma^2})$

(continued on next page)

Table 4 (*continued*).

Reinforcement learning	Bellet (2006)	MDP	It decide the agent action in the partially controlled and partially random environment using π optimized policy	Prediction	Novel MDP	$Q_{opt}(s,a) = \sum_{s'} T(s,a,s')[Reward(s,a,s') + \gamma V_{opt}(s')]$
					Ergodic MDP	$Limited\,Stationary\,Distribution\,d^{\pi}(s) = \sum_{s' \epsilon S} d^{\pi}(s') P_{s's}$
	SARSA (2022)	Q learning	It assesses the Q-value of the next action state from the previous set of actions based on greedy approach	OFF policy prediction	Novel Q	$Q(s_t,a_t) = Q(s_t,a_t) + \alpha(r_{t+1} + \gamma max_a Q(s_{t+1},a) - Q(s_t.a_t))$
					Deep Q	$L_i = (y_i - Q(s_t,a_t,\theta_i))^2 = r_t + \gamma max_a Q(\phi_{t+1},a_{t+1};\theta_{t-1})$
					Double Q	$Q^A_{t+1}(s_t,a_t) = Q^A_t(s_t,a_t) + \alpha_t(s_t,a_t)(r_t + \gamma Q^B_t(s_{t+1}, max_a Q^A_t(s_{t+1},a)) - Q^A_t(s_t.a_t))$ $Q^B_{t+1}(s_t,a_t) = Q^B_t(s_t,a_t) + \alpha_t(s_t,a_t)(r_t + \gamma Q^A_t(s_{t+1}, max_a Q^B_t(s_{t+1},a)) - Q^B_t(s_t.a_t))$
		SARSA	It assesses the Q-value of the next action state from the current set of actions & target policy	ON Policy Prediction	Base approach	$Q(s_t,a_t) = Q(s_t,a_t) + \alpha(r_{t+1} + \gamma Q(s_{t+1},a_{t+1}) - Q(s_t.a_t))$

overview of the recent studies related to fruit/vegetable disease identification, classification, grading, and sorting using machine learning-based methods.

Habib et al. (2020) developed an exploratory online architecture for autonomous disease classification in jackfruit tissue using a ML approach. K-Means grouping-based segmentation method was exploited to segregate the infected region from the non-infected part of the jackfruit image. A linear kernel-based SVM outperformed the other state-of-the-art, achieving a recognition accuracy of 88.67%.

Similarly, ML algorithms can be integrated with optimization algorithms for best feature selection. The optimization algorithms adjust the model hyperparameters and optimize the mean square error between the actual and predicted output. A whale-optimized Artificial Neural Network (ANN) model for tomato disease identification was introduced by Kumar et al. (2020b). The Firefly algorithm was used for efficient morphological-driven color image segmentation, which was further processed by Gray Level Co-Occurrence Matrix (GLCM). El-Dosuky et al. (2020) developed a Cockroach Swam Optimization (CSO) optimized SVM method for apple disease classification. The image features were extracted using the K-means algorithm, which was further classified by CSO optimized SVM method. CSO fine-tuned three attributes (γ representing non–linear hyperplanes, P representing error penalty, and D representing hyperplane splitting the data), thus generating 140 different SVM model variants.

Ji et al. (2018) proposed an apple grading method utilizing the color and surface defects using PSO-optimized SVM. The fruit segmentation from the background noise with color feature extraction was carried out by Otsu's segmentation method for proper H values. Additionally, the external defects boundary of the fruit was identified using the hysteresis threshold canny edge detection method for appropriate I values and segmented utilizing a morphological approach.

SVM and their predecessors play a crucial role in the classification network. Bhargava and Bansal (2020a) presented an SVM-based fruit classification and defect prediction method. The quality of food was ranked into three classes, namely, 1,2, and defective based on the disease severity levels. The experiments revealed that the SVM-based classification outperformed other considered methods showing 98.48% fruit disease classification accuracy. Similarly, Mojumdar and Chakraborty (2021) acknowledged a multi-class SVM for orange tissues disease classification. The image partitioning was performed using K-means clustering, whereas the GLCM method was used for feature extraction. Likewise, Doh et al. (2019) developed an SVM-based method for citrus disease detection. The infected citrus region was segmented using the K-Means algorithm based on four phenotypic attributes, namely, color, structure hole, texture, and physical appearance. The author experimented that SVM out-competed ANN-based methods by showing a mean accuracy of 93.12%.

On the same footprints, Bhargava and Bansal (2020b) introduced an autonomous fruit grading framework to detect the defect of the 'Golden Delicious' apple. Images were segmented using Otsu's and K-means clustering followed by best geometric, statistical & textural feature vector extraction, which were then fed to SVM. Almadhor et al. (2021) developed a framework for disease classification and detection in guava species. The paper experimented with different classifiers (Cubic SVM, Fine KNN, Bagged Tree, Complex Tree, Boosted Tree) for disease level and image level classification. It was concluded that the bagged tree classification method achieved 99.00% accuracy against the healthy guava fruit. El-aziz et al. (2020) implemented a computer vision-based apple disease detection and classification system using SVM and KNN. The proposed system extracted the textual features, whereas, the K-means algorithm was used for segmenting the extracted features.

Furthermore, a low-cost inline tomato grading system was developed by Ireri et al. (2019) using computer vision and machine learning methods. The ROI images were segmented using the average $g - r$ value of the Calyx and Stalk Scar (CS) areas. The paper evaluated the performance of the proposed method and was validated on four state-of-the-art SVM methods, namely, various SVM likewise cubic-SVM (C-SVM), quadratic-SVM (Q-SVM), linear-SVM (L-SVM), and radial basis function-SVM (RBF-SVM), decision tree, ANN, and random forest for different gradings based on the severity. The hybridized RBF-SVM method outperformed other hybrid models by achieving 97.09% accuracy. Yu et al. (2018) fused computer vision and spectral imaging to identify the physical deformation of Nanguo pears. The conventional GLCM method was employed for homogeneity, energy, contrast, entropy, and correlation discriminative texture feature vectors, which were tested against SVM and backpropagation networks.

Additionally, the segmentation technique helps to partition the images into meaningful ROI, where each region represents a distinct object. Dubey et al. (2020) implemented a fast and effective disease recognition framework using SVM. The resized grayscale images were carried out for infected area segmentation using canny edge detection to classify fruitage in defective and non-defective binary and first, second & final defect multi-classification. Additionally, unsupervised learning methods can be utilized for assembling the defective area. The research paper showed that the Gaussian Mixer Model (GMM) segmentation method overshadows the other image processing-based segmentation method by showing a silhouette score of 0.63 (Roy et al., 2021b).

Clustering, an unsupervised learning method, divides the data into clusters where data points are more similar to one another than to those in other clusters. Kumari et al. (2021) presented a backpropagation-based mango grading method. The Extended Fuzzy K-Means grouping algorithm was employed for detecting infected segmented areas, which experimentally performed better than Fuzzy C-Means and traditional K-Means clustering. The color, texture, and geometric features were optimally extracted using Maximally Correlated Principal Component Analysis (MCPCA). Lastly, severely infected regions were classified by

Table 5
Machine learning fruit & vegetable disease detection techniques.

Author	Research methodology	Species (Approx)	Application or type of disease	Classification accuracy	Findings
Habib et al. (2020)	K-Means and Linear SVM	360 jackfruits	Classification of bad and good quality tissues	88.67% (Accuracy), 71.67% (Precision & Recall), 92.92% (Specificity)	The lightweight real-time model with more testing samples should be considered.
Kumar et al. (2020b)	Firefly algorithm, Whale Optimization based ANN	900 tomatoes	Disease (ripe, unripe, blotchy ripe and sour rot) and healthy	94.11% (accuracy), 96.75% (sensitivity)	Deep learning approaches should be combined with optimization methods to sustain the better results
El-Dosuky et al. (2020)	K-means, CSO with SVM classification	320 apples	Rot, scab, and blotch	94.65%	Noisy data with large samples should be further added for better performance.
Bhargava and Bansal (2020a)	Sparse representative, SVM, ANN, KNN	4359 apples, 918 avocados, 3805 bananas, 3050 oranges	Binary Classification (fresh and rotten)	98.48% (Accuracy), 95.72% (Defected maximum accuracy)	Future approaches include more significant segmentation features. Also, deep belief network can be combined with meta-heuristics optimization methods
Mojumdar and Chakraborty (2021)	K-means, GLCM, SVM	517 many fruitage variants	Binary classification(healthy & unhealthy)	82.30%	Clustering outlier data should be emphasized
Doh et al. (2019)	K-means, ANN, SVM	120 citrus	Scab, greening, melanose, canker, black spot, anthracnose	93.12%	Future research include data augmentation to develop dataset diversity
Bhargava and Bansal (2020b)	Otsu's, K-means, SVM	212 apples	Detection of healthy and defected datasets	96.81% and 93.00% with given two datasets	Other SVM variants and their hybridization should be focussed
Almadhor et al. (2021)	Cubic SVM, Fine KNN, Bagged Tree, Complex Tree, Boosted Tree	393 guava species	Rust, canker, dot and mummification, and healthy guava	99.00%	The local minima of the SVM and their variants can lead to future research hotspot
El-aziz et al. (2020)	K-means, SVM, KNN	320 apples	binary (healthy/rotten apple)multi-level (blotch, rot, scab)	97.90%	Augmentation techniques & weight optimization should be explored for data divergence
Ireri et al. (2019)	Calyr and stalk scar detection, SVM, decision tree, ANN, RF	8000 tomatoes	Healthy and defected tomatoes of various grading	97.09%	The performance degrades as the number of grading class increase, leads for future work
Yu et al. (2018)	Backpropagation, SVM	300 nanguo pears	External deformation with four classification	normal (90.00), rotten(100.00), deformed(80.00), insects infected(100.00), rust infected(90.00)	The future work can emphasize on the internal defects of the agro-products
Ji et al. (2018)	Otsu's, canny edge detection, Particle Swarm optimized SVM	200 apples	Identification of multiple grading of apple	91.00%	More complex high dimensional data can be further explored
Dubey et al. (2020)	Canny edge detection, SVM	500 apples, 500 pears, 500 pomegranates	Healthy, and defected images which are further categorized to first, second and third infection level	SVM model shows 98.50% over GoogleNet(79.40%) and KNN(89.40%)	More feature selection may lead to increase in accuracy rate
Kumari et al. (2021)	Fuzzy K-means, GLCM, PCA, backpropagation	4500 mangoes	ripe,unripe, & defected	98.40%, 98.20%, 97.20% for ripe, unripe & defect classes	Principal components can be explored for high dimension data.
James and Sujatha (2021)	Combination of K-Means and feed forward back propagation neural network.	500 apples	10 different diseases	98.10%	Generative adversarial network should be considered for increasing the false detection rate
Roy et al. (2021b)	Histogram, random walker and gaussian mixer	5031 apples, 4716 bananas	Two species with sub-classes rotten and fresh	0.63(silhouette score)	More features may overfit the GMM model, should be reviewed
Kumar et al. (2020a)	Cascade of two SVM along with Gabor wavelet transfer	1250 tomatoes	Tomato or no tomato with defective or non-defective class & its disease type	97.74% (Accuracy), 99.38% (Specificity), 96.55% (Sensitivity), 98.24% (Precision)	The feature extraction for other defects like anthracnose, mold, rot should be analyzed
Hemamalini et al. (2022)	K-means, different classifiers(SVM, KNN, C 4,5)	113 apples, 137 mangoes	Determination of fresh and rotten class	98.00% (accuracy), 91.00% (sensitivity), 99.00% (specificity)	Best classifier can be incorporated with robots for fresh fruitage selection

(*continued on next page*)

the Backpropagation Based Discriminant Classifier (BBDC) into three categories (good, average, and bad)

Analogously, James and Sujatha (2021) implemented a web-based Hybrid Neural clustering method that exploited the feed-forward neural

Table 5 (*continued*).

Al-Sanabani et al. (2019)	PLS regression and SVM regression	80 mangoes	Identification of freshness levels of different mangoes types	coefficient of determination 0.96(training) and 0.95 (testing)	The result may sensitive to the initial data points, may be examined
Sugiarti et al. (2021)	K-means, GLCM, Naïve Bayes	391 apples	rot, scab and blotch	96.43%	The zero-frequency problem can be further investigated
Ghodke et al. (2022)	LSVR, FFC	300 four different fruitage tissues	Binary classification	87.00% (sensitivity)	Various variant of SVM can be further investigated
Mohammed et al. (2023)	LR, DFR, Azure Ml studio	100 dates with Khalas and Barhee	Date Palm Mite	$R^2 = 0.84$ (meteorological), $R^2 = 0.89$ (physicochemical), $R^2 = 0.92$ (hybridization)	Real-field application deployment for other mite genres should be focussed for research work.
Dubey and Rocha (2023)	Two-class jungle decision method, SVM, naive bayes, decision tree, LR, NN	1000 pears	Calcium deficit and non-deficit pears	98.00%	The model can be further extended for multi-class nutrient deficit labels for multiple fruits
Tripathi and Maktedar (2023)	Patch-based PCA for dimension reduction and ensemble classifier(RF, SVM, ANN, KNN)	540 mangoes	Three grade qualities (healthy ripe, unripe & disease)	93.33%	The model hyperparameters can be further reduced for real-time applications

network and K-Means clustering approach for training and testing the apple disease classification. The proposed algorithm segmented the diseased part using the Maximum Likelihood Pixel Fusion method and feature extraction using the Essential Factor Investigator (EFI) method. Kumar et al. (2020a) devised a dual classification system using cascading of two SVM models for tomato disease classification. In the first stage, binary classification was performed to determine tomato and non-tomato species using species feature vectors designed from the captured images. The color features were used to classify the tomato quality based on the infectious level. The infected area was then segmented through the Gabor wavelet transform for good or bad binary classification, which was then cascaded to SVM for classifying the disease category. In a similar manner, an SVM-based method for mango and apple disease identification was introduced by Hemamalini et al. (2022). The k-means-based segmentation was performed to detect the infectious area using color and texture features. The experimental results affirmed that the developed SVM-based method outperformed other considered methods by achieving 98.00% accuracy.

Furthermore, the regression algorithms allow the ML models to map input variables to continuous output for fruit disease detection. Al-Sanabani et al. (2019) identified the freshness level of multiple cultivars of mangoes by comparing the Partial Least Square (PLS) & SVM regression models. The collected Near Infrared (NIR) images were calibrated using extended multiplicative signal correction and the Gaussian method for noise removal, which were further processed using an MA871 optical machine (Refractometers, 2022) for refractive index measurement to determine the Brix sugar concentration. Similarly, the Logistic Support Vector Regression (LSVR) based sorting method for six fruits and vegetables, namely, bananas, apples, lime, orange, pomegranate, and guava was implemented by Ghodke et al. (2022). The shape features were extracted using the Fast Fourier Convert (FFC) method, which became fed to the LSVR classifier into low and high-quality fruits for binary disease classification.

Sugiarti et al. (2021) investigated an image processing-driven solution for disease recognition in apple fruits. The image samples were segmented to identify the ROI containing the infected part using the k-means grouping method, which was further fed to the GLCM technique to obtain homogeneity, energy, contrast, and correlation discriminative texture feature vectors. The experimental results witnessed that the developed naive-bayes classifier surpassed other considered methods by achieving the disease classification accuracy of 96.43%. Mohammed et al. (2023) designed the autonomous palm mite detection in two cultivars, namely Khalas and Barhee, of the date fruits. Decision Forest Regression (DFR) and Linear Regression (LR) models were used to extract the meteorological and physicochemical features. The experiments showed the developed DFR model showed better accuracy with $R^2 = 0.91$.

Furthermore, a non-destructive nutrient detection for pears was proposed using a two-class decision jungle network (Dubey and Rocha, 2023). The geometrical features, namely, area, perimeter, eccentricity, diameter, major and minor axis, and orientation, were retrieved from the dataset and classified into calcium-deficient and healthy pears. The proposed method achieved 98.00% accuracy. Tripathi and Maktedar (2023) introduced an ensemble classifier that utilized random forest, SVM, KNN, and Lion-Binary crossover mask base Whale Optimized (LBWO) ANN for automatic mango grading. The RGB images were denoised using the adaptive Gaussian method. Gray Level Run-Length Matrix (GLRM), NIR, and GLCM features were extracted from the processed dataset. The extracted feature vectors were reduced to the principal vectors using the PCA method, which was then fed to the ensembled network for disease classification.

5. Deep learning

The study of infectious and non-infectious pathogenic agent traits is the foundation for agro-product disease prognosis and epidemiology. The factors affecting fruit and vegetable symptoms may be in the context of environmental conditions, climate conditions, accessibility of pathogens, and many more. The extensive methodology for disease diagnosis is to identify the fruitage lesion followed by pre-processing and detection (Moore et al., 2001). Classical machine learning and deep learning are two mainstream methods in precision agriculture. Recently, Deep Learning (DL) techniques have been significantly used by researchers for the early identification of fruit & vegetable diseases due to automatic feature extraction. The DL techniques have been revolutionized and recognized after winning the 2012 ImageNet Large Scale Visual Recognition Challenge (ILSVRC) (ImageNet, 2023) by AlexNet architecture (Krizhevsky et al., 2012) over traditional SVM (Cortes and Vapnik, 1995). The winning architectures in the successive years are ZFNet, GoogLeNet(InceptionV1), VGGNet, ResNet, ResNeXt, SENet, and PNASNet-5, respectively. The top-5 error rate has been reduced from 15.3% to 3.8%, which proves the direct correlation between classification accuracy and model depth (He et al., 2015).

5.1. Fruit & vegetable disease detection using DL techniques

The nutritional content of fruits and vegetables has boosted the researchers to mine the precision agriculture methods by appending more depth to the pre-trained deep models. In the last decade, DL models have received significant attention from researchers and agronomists due to their ability to automate feature extraction and classification of images with minute regions of interest. Table 6 outlines the recent research work carried out in the fruitage domain exploiting DL methods.

Janarthan et al. (2020) designed a region-based lightweight deep model for classifying citrus diseases. The proposed model consisted of three major components: deep siamese embedding, clustering module, and neural architecture. The deep embedding module calculated the patch embeddings of all eligible patches using Euclidean distance, which were grouped using the K-means clustering. Dhiman et al. (2022a) proposed the classification model to detect the low, medium, high, and healthy severity levels of citrus. The images were segmented by the felzenszwalb graph method, whereas the Local Binary Pattern (LBP) method was used to cluster similar regions. The highest similarity index regions were merged to determine the Intersection over Union (IoU) regions which became an input to the Adam-optimized VGGNet model.

The optimization methods can be incorporated with deep belief networks to optimize the hyperparameters. Rehman et al. (2022) presented a WAO and MobileNetV2 & DenseNet201 hybrid model for citrus disease detection. The image dataset was expanded using geometric augmentation, followed by an average-based threshold contrast stretching for image quality enhancement. The classification was performed using five classifiers namely, three state-of-the-art SVM classifiers, ensemble subspace discriminant, and LDA. Likewise, Tian et al. (2021) utilized the different network depths to better classify apple disease. The optimized InceptionV4 and Inception-ResNetV2 models were concatenated with the multi-scale framework (Szegedy et al., 2015). An improved You Only Look Once (YOLO) model was leveraged for identifying and computing the number of olive fruit flies (Mamdouh and Khattab, 2021). The author exploited the negative training sample images by appending original Dacus Image Recognition Toolkit (DIRT) images with Leeds butterfly datasets (LEEDS, 2022). The images were normalized using the yellow average color method to unify the trap background color.

An improved U-Net architecture was presented by Roy et al. (2021a). It enhanced the U-Net architecture by incorporating a convolution2D, max pooling 2D, and convolution transpose 2D layer between the encoder and decoder. The depth level in the suggested En-UNet architecture was increased from 128 to 512, resulting in increased disease classification accuracy. Yasmeen et al. (2021) introduced the InceptionV3 and ResNet-18 models for citrus feature extraction, which are optimized by an Improved Genetic Algorithm (ImGA). The disease identification was performed using ten state-of-the-art methods. Experimental results demonstrated that the F-KNN method surpassed the other remaining considered methods in terms of accuracy and F-measure. Yao et al. (2022) presented a mask-scoring R-CNN model where the loss function of the original R-CNN model was replaced with the focal loss to reduce classification error and class imbalance problems. An improved instance segmentation method was used to identify, locate, and segment the lesion for prominent regions from the 21 classes of the peach.

Consequently, an improved YOLO-JD architecture was suggested for real-time disease detection in jute (Li et al., 2022). The Spatial Pyramid Pooling Module (SPPM), Deep Sand Clock Feature Extraction Module (DSCFEM), and Sand Clock Feature Extraction Module (SCFEM) were used for high-quality feature extraction. Lastly, the final detection module, in combination with the Non-Max Suppression (NMS) method, was used to select the final anchor box. Oppenheim et al. (2019) addressed the disease identification problem in potato tubers using deeply improved CNN architecture. The resized images were classified using a deep CNN model with eight tunable layers, which includes five convolution2D with max pool layers, three fully connected layers, and one softmax activation layer.

A grading and sorting method was developed by da Costa et al. (2020). The method utilized the fomesa machine to capture 1500 tomato images. Vasumathi and Kamarasan (2021) classified the pomegranate fruit into infected or non-infected classes utilizing the deep convolution network pipelined with long short-term memory for detection. Selvaraj et al. (2019) collected 12,600 balanced real-world banana images at different growing phases with different resolutions, illumination conditions, and varying environmental conditions to avoid underfitting, overfitting, and generalization errors. The author performed the analytical study of Faster R-CNN InceptionV2, Faster R-CNN ResNet50, and SSD MobileNetV1 networks. The experiments were conducted to detect disease and pest symptoms and classify the whole plant, fruit bunch, pseudo-stem, and cut fruit. Meshram et al. (2021) compared the misclassification rate of two proposed models FC_InceptionV3 (Fruit Classification InceptionV3) and MFC_InceptionV3 (Merged Fruit Classification InceptionV3) by utilizing the "Divide and Conquer" concept (The design, 2022). FC_InceptionV3 architecture implemented the pre-trained model InceptionV3 while MFC_Inception V3 architecture was developed using a novel Merged Net (MNet) framework. The first FC_InceptionV3 architecture dealt with multiple fruit recognition, while the second FC_Inception V3 architecture dealt with fruit quality bad or good binary classification simultaneously, which were merged together into a single pertinent classification class.

Although some researchers reviewed the state-of-the-art models for automatic disease detection, Thangaraj et al. (2020) worked on the state-of-the-art Adam-optimized MobileNet model using transfer learning for automatic disease recognition in avocados. The transfer learning method replaces the final classification layer with a fine-tuned layer with eight disease classes. Kizilboga et al. (2020) manifested AlexNet and improved CNN deep neural networks to classify the disease in the apple and quince datasets. Consequently, Saini et al. (2021) studied the pre-trained architectures, namely, VGG16, ResNet-50, InceptionV3, and VGG19, using the transfer learning method for identifying and classifying the diseases in citrus. The author experimentally showed that the VGG19 prototype outperformed the rest state-of-the-art models with a classification accuracy of 99.89%.

A novel Correlation Coefficient & Deep Features (CCDF) method was adopted for disease detection in multiple fruits & vegetables (Khan et al., 2018). The first stage performed a contrast stretch operation for better infected area visualization. The second stage exploited the Caffe AlexNet and VGG16 state-of-the-art models for feature extraction, which were fused together to aggregate the features. The Genetic Algorithm (GA) selected the most critical features from the maximum feature vectors, which became the input to the multi-class SVM classifier.

Buyukarikan and Ulker (2022) compared the amalgam of VGG16 and VGG19 for 4096 feature extraction, ResNet50, ResNet152, and ResNet101 for 2048 feature extraction, and LR, SVM, XGBoost, and RF models with fivefold cross-validation approach for disease classification against original (1080) and augmented (4320) apple data images. Furthermore, Turkoglu et al. (2019) applied three tier Long Short Term Memory (LSTM) model as an aggregated majority voting classifier for multi-class disease classification in the apples. DenseNet201, GoogleNet, and AlexNet extracted the feature vectors of dimensions 1000, 1024, and 4096, respectively, which were then taken as an input to three LSTM layers followed by the combination of an FC, softmax, and a classification layer. Khattak et al. (2021) addressed the automatic citrus species disease classification problem by integrating manifold CNN layers. The proposed model exploited the first convolution and max pooling layers for feature extraction and dimension reduction, respectively. On the other hand, the second convolution and max pooling layers were used for high-level feature mapping and image downsampling. Finally, the flatten layer is followed by the classification layer for disease detection.

An autonomous multi-fruit sorting system was designed by Dhiman et al. (2021). The grayscale image quality was enhanced using contrast

Table 6
Deep learning fruit & vegetable disease detection techniques.

Author	Research methodology	Species (Approx)	Application/Type of disease	Classification accuracy	Findings
Janarthan et al. (2020)	Model-128 & Model-112 CNN & K-Means	609 citrus tissues	Canker, black spot, scab,melanose and greening	95.04%	Model with less memory utilization can be chosen for parameter tuning
Dhiman et al. (2022a)	Improved VGGNet-16 with transfer learning	3309 citrus	Severity levels (low, medium, high, and healthy)	99.00%, 97.00%, 98.00%, 96.00%	The model may be deployed on real-time citrus images with complex background
Rehman et al. (2022)	Whale optimized MobileNetv2 & DenseNet201 & 5 classifiers	3023 citrus tissues	Canker, black spot, anthracnose, scab, melanose, greening, and healthy	95.70%	Data replication in DenseNet should be focused with other improved feature selection methods.
Roy et al. (2021a)	Enhanced Unet(En-Unet) with semantic segmentation	4035 apples	Rotten and fresh apple	97.54%	Improvise the model to perform the multi-class classification
Yasmeen et al. (2021)	GA optimized ResNet18 & Inceptionv3 & SVM	7500 citrus tissues	Canker, black spot, anthracnose, scab, melanose, greening and healthy	97.70% (Fruit), 94.00% (Leaves), 99.50% (Hybrid)	Discussion on premature convergence issue should be taken into account.
Tian et al. (2021)	CycleGAN augmented multiscale dense Inceptionv4 & ResNetv2	5801 apple tissues	Recognition of 11 categories of various forms of anthracnose, scab, rust, and ring rot disease	94.31% (Inceptionv4), 94.74% (Inceptionv2)	Cross-domain image translation for image expansion led to future work
Mamdouh and Khattab (2021)	Modified YOLOV4	1680 olive fruit flies	Detection and counting of olive fruit flies	84.00%, F1-Score (0.90), mAP (96.68%)	Design of multi-class classifier (male or female Olive flies) & augmenting the negative training samples that mimic the original olive fly
Yao et al. (2022)	Mask R-CNN and Scoring R-CNN	1560 peach tissues	Brown rot, anthracnose, bacterial shot hole, powdery mildew, gummosis,scap, and curl	0.46 (Segment ation Accuracy)	The model can be extended for video data for temporal information
Li et al. (2022)	YOLO-JD architecture with SPPM, DSCFEM, and SCFEM feature extraction	4418 jute	Soft rot, dieback, jute chlorosis, jute mosaic, anthracnose, stem rot, top blight, black band and two pests (Comophila sabulifers & Hairy Caterpillar)	96.63% (mAP) and 95.83% (F1-Score)	Improved YOLO-JD should be deployed on handheld devices
Oppenheim et al. (2019)	Pre-trained CNN Architectures	2465 potatoes	Common scab, black scurf, silver scurf, black dot, and disease free patches	96.00% for 90:10 dataset	Improvise the models with diversity and variability of the data
da Costa et al. (2020)	ResNet50 with L2 regularization	43 843 tomatoes	Detection of normal and abnormal Tomato	94.60%	Model can be further extended by considering high variance for bias
Vasumathi and Kamarasan (2021)	CNN, LSTM	6519 pomegranates	Healthy and unhealthy pomegranate	98.17%	Diseases like heart rot, bacterial blight, and cercospora can be further identified
Selvaraj et al. (2019)	MobileNetv1 with transfer learning	12 600 banana tissues	Fusarium Wilt of Banana, Banana Bunchy Top Disease, Black Sigatoka, Xanthomonas Xilt of Banana, Yellow Sigatoka, and Banana Corm Weevil	99.72%	Handheld application for disease detection may be future work
Wang et al. (2019)	Study of VGG16, ResNet-50, ResNet-101, MobileNet with Faster R-CNN and Mask R-CNN	286 tomatoes	Malformed, anthracnose, blotchy ripening, ulcer, puffy, gray mold, dehiscent, virus,blossom-end rot, sunscald	99.64% (Mask R-CNN), 88.53% (Faster R-CNN)	The model can be investigated on low-resolution images and other variety of crops
Tian et al. (2019)	CycleGAN augmented DenseNet YOLOV3	640 apples	Detection of anthracnose disease	0.81 F1-Score, & 0.032 s detection time	Future direction can lead to consider those affected regions which are not visible by naked eyes or extremely unclear
Meshram et al. (2021)	Model FC_Inceptionv3 & MFC_Inceptionv3 using Inceptionv3	1200(apples, bananas, guavas, limes, oranges, pomegranates)	Detection of Binary classification either Good or Bad	99.92%	Future direction can focus on the large computation power and training time of Inceptionv3 network
Han et al. (2021)	Improved Mask R-CNN	74 apples	Detection and localization of anthracnose disease	72.26% (mAP)	Future study should be followed for extensive edge segmentation.
Thangaraj et al. (2020)	Adam optimized mobileNet model	9000 avocado images	Anthracnose, sun bloch, cercospora, stem end rot, phytophthora, seed moth, rat bite, scab	96.82%	Slow convergence rate of the Adam optimizer needs to be focused

(continued on next page)

Table 6 (*continued*).

Kizilboga et al. (2020)	AlexNet	600 apples and quinces	Black spot, powdery mildew, worm, alternaria and quince diseases like blue mold rot and brown rot	83.30%	Dataset can be expanded to include the real-time images to sustain the reliability of the model
Khan et al. (2018)	GA optimized Caffe-Alexnet network and SVM	2700(apples and banana tissues)	Scab, rot and banana disease like diamond spot, cordial, deightoniella, and sigotka	98.60%	Feature extraction and segmentation domain can be further explored to get better accuracy
Saini et al. (2021)	Study of VGG16, Resnet-50, InceptionV3, and VGG19 using transfer learning	705 citrus	Canker, black spot, anthracnose, scab, melanosis, greening, powdery mildew and healthy	99.89%	Augment the dataset by merging the real-time images to preserve the reliability of the model
Buyukarikan and Ulker (2022)	VGG16 with SVM, RF, KNN, XGBoost	1080 apples	Shriveling, superficial scald and bitter pit in granny smith, golden delicious images	96.11% (data-1), 96.09% (data-2)	The training time and parameters needs to be reduced for future work
Turkoglu et al. (2019)	GoogleNet, AlexNet, DenseNet201 and LSTM	1192 apples	Monilinia laxa, aphis pomi deg, venturia inequalis and eriosoma lanigerum	99.20%	Multifruit disease classification led to the future research work
Khattak et al. (2021)	Conv layer-1, Maxpool-1, Conv-layer2, Maxpool-2, Flatten layer	1067 citrus tissues	canker, black spot, melanosis, greening and scab	95.65%	Hybridization of CNN, RNN, LSTM, and Bi-LSTM should be reviewed
Dhiman et al. (2021)	CLAHE, Canny edge detection, PCA & RNN	4000 multiple fruit varieties	Bad and good quality	98.47% (Accuracy), 98.93% (Precision), 1.53% (MSE)	New deep learning model on a multi-fruit system with dynamic feature extraction should be enlightened
Chaturvedi et al. (2023)	ResNet50, EffcientNetB4, VGG19, Inceptionv3, and DenseNet201 using transfer learning	43 843 tomatoes	Aberrations, rotten, bruises, and cuts	97.97% (EffcientNetB4)	The pre-trained models are susceptible to data complexity and compatibility issues
Dhiman et al. (2023)	Fusion of CNN & LSTM, magnitude-based pruning and nonpruning	2950 citrus	canker, black spot,Melanose, greening and scab	97.18% & 98.25%	The model size can be further decreased using post-quantization method
Hu et al. (2023)	YOLOV5 pre-trained model with attention block, DIoU as modified loss function	845 citrus	Binary classification (Defect1, Defect2)	95.50% (mAP), 94.00% (Precision), 95.10% (Recall)	Spectral imaging techniques can be further deployed to study the internal defects

enhancement and CLAHE edge operator for light balancing and object boundary detection. Using the PCA method, the author proposed a recurrent neural network for binary classification after extracting good-quality and bad-quality feature vectors. Moreover, Chaturvedi et al. (2023) presented autonomous external lesion detection in tomatoes. The paper compared the five state-of-the-art architectures. The experimental results witnessed that EfficientNetB4 surpassed the other models by achieving the recognition accuracy of 97.97%.

A CNN-LSTM-based model was used for early disease classification of the citrus (Dhiman et al., 2023). The images were segmented using canny edge detection and the K-Means clustering. The model was trained and tested with pruning and quantization methods for five-class classification. Moreover, an improved YOLOV5 method was proposed for two defect detection, namely black spots and surface damage in citrus (Hu et al., 2023). The Convolution Block Attention Module (CBAM) was integrated into the YOLOV5 model, whereas the loss function was optimized using Distance Intersection over Union (DIoU). The performance was evaluated using Generalised Intersection over Union (GIoU), Complete Intersection over Union (CIoU), and Efficient Intersection over Union (EIoU) loss functions.

6. Internet-of-things and Cyberphysical System

Recently, Internet-of-Things (IoT) and Cyberphysical Systems (CPS) have emerged as innovative and promising edge-cutting technologies, offering trustworthy and effective solutions for smart agriculture. IoT comprises heterogeneously connected physical entities, empowering them with new autonomous computation power. Moreover, IoT offers new capabilities, connecting virtual and physical devices, where Machine-to-Machine (M2M) interaction makes the interconnection between things and cloud applications. The Internet Architecture Board (IAB) defines IoT as a collection of embedded network devices utilizing communication internet protocols. In 1999, Kevin Ashton first referred to IoT as distinctly identifiable, coherent, interconnected entities with Radio Frequency Identification (RFID) technology to track and count goods in supply chains (Floerkemeier and Lampe, 2005). The Indian government has launched the "Digital India" flagship program intending to incorporate digitalization into every aspect of society. Different research organizations have predicted a wide span of projections concerning the significant impact of IoT on the future internet and their global economy. For instance, Cisco (CISCO, 2022) expects its global economy of almost $24 billion in IoT connections. On the other side, Huawei (Huawei, 2022) projects $100 billion in network-connected devices, and McKinsey Global Institute (Manyika et al., 2015) forecasts their IoT global economy from $3.9 to $11.1 trillion by 2025. Thus, internet traffic will exponentially grow as connected IoT devices increase. Cisco (CISCO, 2022) expects the volume of internet traffic from non-PC devices will rise from 40% to 70%. The volume of "Machine to Machine" (M2M) interconnections in all IoT verticals, including farming, healthcare, industrial, smart city, automotive, transportation, home, unmanned aerial vehicles, and others, will rise from 24% to 43% in the coming future. As per the global precision agriculture market trends (Agriculture, 2022), it is expected to touch $15.34 billion by the end of 2025.

6.1. Iot enabled E-Agriculture

IoT-based precision farming comprises multiple physical components (sensors, actuators, microcontrollers), data gathering from IoT-based sensors, data processing, and data analytics. A sensor (transducer) is a physical device that senses the abrupt change in the environment and converts the data from physical phenomena (like humidity, pressure, temperature, light, and motion) to another form of energy. Conversely, an actuator is a physical device that performs in the counter

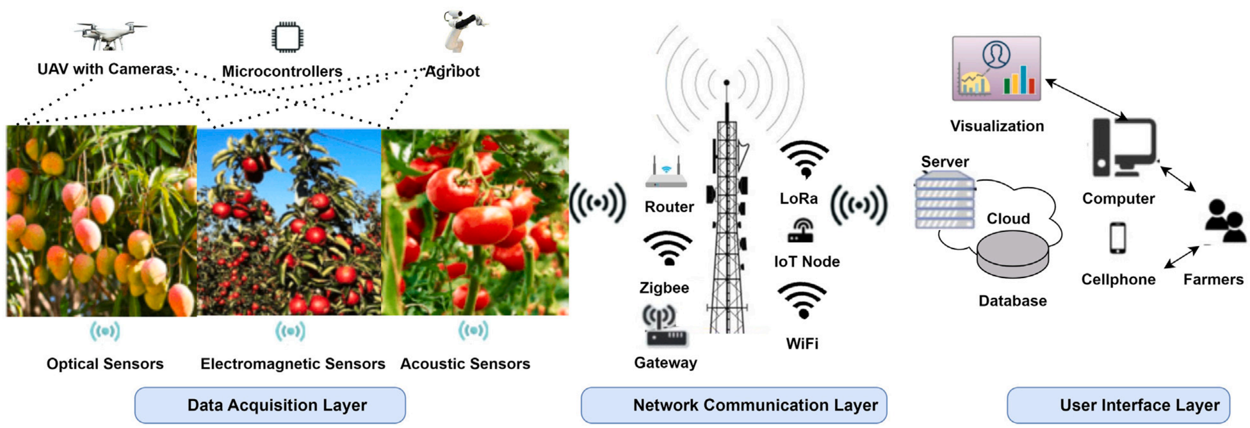

Fig. 4. Flowchart of IoT enabled E-Agriculture.

Table 7
IoT development board.

Architecture	Microcontroller	Memory	Clock speed (MHz)	I/O
Arduino Nano	ATmega328P	32 KB	16	14 (Digital), 6 (Analog)
Arduino Uno	ATmega328	256 KB	16	14 (Digital), 6 (Analog)
Arduino Mega	ATmega1280, ATmega2560	256 KB	16	54 (Digital), 16 (Analog)
ARM	LPC2148	32-512 KB	60	32 I/O
ESP2866	RISC	4 MB	80	16 (Digital), 1 (Analog)
Raspberry Pi	Broadcom BCM2387 chipset	1 GB	1.2 GHz	16 GPIO (PI 1), 40 GPIO (PI2& 3)

direction of the sensor. Whereas, microcontroller devices (ATMEGA series, Arduino, Raspberry Pi) control the functionality of device identification and service discovery in the network through network-connected remote devices. To facilitate the transceiving of agriculture data, different application layer messaging protocols like WebSocket, Message Queue Telemetry Transport (MQTT), Secure MQTT (SMQTT), Extensible Messaging and Presence Protocol (XMPP), Advanced Message Queuing Protocol (AMQP), Representational State Transfer (REST), Data Distribution Service (DDS), and Java Message Service (JMS) and communication protocols like Zigbee, IPv6 over Low Power Personal Area Networks (6LoWPANs), Low Power Wide Area Network (LPWAN), Bluetooth Low Energy (BLE), Z-Wave, IEEE 802.15.4, SigFox, Wireless HART are employed to provide homogeneous and standard communication from different embedded devices. IoT edge or ubiquitous fog computing can be analyzed in various domains of precision agriculture like soil monitoring, weather prediction, livestock management, field monitoring, greenhouse monitoring, and pest monitoring fields. Moreover, Agriculture robots (agribots) can be used to maximize crop yields and resource utilization using innovative advanced Information and Communication Technology (ICTs) (Gollakota and Srinivas, 2011). E-agriculture involves gathering real-time data from the IoT sensors, monitoring the fruitage environmental attributes, routing the data from the farmlands to the control center for processing the collected information, and presenting the result visualizations to the agrarians. The basic flowchart of the IoT-enabled E-Agriculture has been depicted in Fig. 4, which has been layerwise discussed below:

- **Data Acquisition**: The initial layer is the data sensing layer, which comprises various smart sensors to gather real-time information from the farmlands. For instance, optical sensors collect the reflectance data and measure soil attributes like moisture, and pH. These smart sensors are mounted at a specific height from the farmland's surface to observe and regulate diverse physiological indicators related to the soil, fruitage crops, and environment, which are either directly or indirectly crucial to disease identification in the fruitage domain. Additionally, UAVs equipped with thermal cameras, IoT cameras, and agribots are employed in the agriculture fields to measure the ecological factors for disease classification, displaying the findings through images.
- **Network Communication**: The layer focuses on the connectivity between the sensory nodes and data routing technologies. Wireless Sensor Networks (WSNs) enable real-time monitoring & enhancement of fruit production quality, alongside the capability for extensive monitoring with a high density of data points. It can ensure the optimal conditions for the highest fruit yield by continuously tracking the complex interaction between the IoT sensor nodes distributed across the field using communication protocols. For instance, Long Range (LoRa) communication protocol facilitates long-range transmission with minimal power consumption, ideally suited for battery-powered wireless IoT devices. It employs the Decision Support System (DSS) to analyze the aggregated data using micro-controllers like Arduino, Raspberry Pi, and ESP8266, as shown in Table 7.
- **User Interface**: This layer integrates the smart prediction models to deliver visualization insights concerning disease identification in fruits and vegetables. It presents the informative decisions collected from different IoT devices in an understandable format. It guarantees that both informational and control features about the farms are readily available on diverse smart devices, such as smartphones, tablets, computers, and specific interfaces. This accessibility allows farmers to engage with the IoT system at their convenience from any location. The end-user agrarians can take immediate action based on the IoT system's response for higher production yield.

6.2. Fruit & vegetable disease detection using IoT

In the last decade, IoT has significantly placed in fruit disease classification and prediction by employing deep learning, machine learning, and computer vision techniques. Table 8 contextualizes the study of existing IoT-integrated approaches for autonomous visual disease detection systems of real-time fruits and vegetables.

Zhang et al. (2022) integrated the IoT ecosystem with multi-scale dual-channel CNN (DMCNN) to identify apple diseases. The IoT architecture consisted of Jetson Xavier Nx and TensorRT accelerator. The method extracted relevant texture information using twin-factor weight optimization in the attention algorithm. The extracted features

Table 8
Internet-of-things fruit & vegetable disease detection techniques.

Author	Research methodology	Species (Approx)	Application/Type of disease	Classification accuracy	Findings
Zhang et al. (2022)	8 GB Jetson Xavier NX edge node with 5.00 MP CSI camera & TensorRT accelerator, DMCNN network	2330 apples	Ring rot, anthracnose and healthy	98.50%	Further invention can be done to reduce the parameters for feature fusion to speed up the rate
Mishra et al. (2019)	An infrared camera, Arduino processor, a wifi module, and mobile device.	5 fruitage types	Ripening or unripened	100.00%	Dataset can be augmented to sustain the reliability of the model
Zhou et al. (2021)	UAV and near ground digital images acquisition, images normalization using calibrated reflectance panel and gray card and YOLOV3 architecture	5843 strawberries	7 maturity stage for near ground images and 3 maturity classes for UAV images	93.00% (for UAV images), 94.00% (near ground images)	High-resolution UAV images can be further explored to classify more classes of strawberry
Akhter and Sofi (2021)	Six sensors comprising of IRIS incorporated with an MTS 420 sensor board and MIB 520 gateway as wireless sensor/IoT nodes and LR model	265 770 apples	Scab	Error of 9.30, 0.13, 0.17 for Intercept, temperature & wetting duration	Further research can be investigated on the interoperability, security, scalability, and configuration of network
Jawade et al. (2020)	Atmega microcontroller board, humidity and temperature sensors from on-field IOT kit, Thingspeak cloud storage platform, RF classifier	mangoes	Forecasting of disease on Alphonso mangoes	Least mean absolute error of 0.046	Forecasting result can be further improved by incorporating deep belief convolution network
Ismail and Malik (2022)	2 Raspberry Pi module (Google AIY Vision Kit and module having touchscreen) and study of the performance of various models like EfficientNet, MobileNetV2, ResNet, NASNet and DenseNet.	9091 (apples, bananas)	healthy (ripe) and unhealthy(overripe) of apple(banana)	93.80% (Banana), 96.70% (Apple)	The model can be further enhanced by mitigating image pre-processing and segmentation approaches
Arya and Gangwar (2021)	Sensors for various environmental parameters, internet, gateways, wi-fi for communication, OBRB for feature extraction and RANSAC for feature mapping	13 599 (apples, bananas, and oranges)	Fruit freshness level	97.50%	The upper time bound of the algorithm can be further limited with optimal solutions.
Aasha Nandhini et al. (2018)	Raspberry Pi 3 as the sensor platform and Thingspeak platform, GLCM feature extraction, compressed and uncompressed with CS and OMP, and SVM classifier	Multiple fruitage cultivars	Anthracnose, blight, spot, alternaria	98.51%	The accuracy of the proposed algorithm can be further validated by stable convergence rate of the basis pursuit
Bhole et al. (2020)	GLCM feature extraction, KNN and RF classifiers for thermal images	Multiple fruits & vegetables	Binary class(non-infected & infected)	97.37% (RGB), 84.97% (Thermal)	Thermal images fail to identify images with erratic temperature
Gawas et al. (2021)	The microcontrollers (MQ3 & MQ4 sensors) for ethanol and methane concentration, three cameras, nodeMCU8266, and advanced InceptionV3 model	13 599 (apples, bananas, and oranges)	Binary classification(fresh and rotten)	99.17%	Future possibility is real-time images that are more disease prone
Pawara et al. (2018)	Humidity sensor(AM2302), temperature (DHT22), Arduino Uno microcontroller, ThinkSpeak cloud, SIM800 Quad band GSM/GPRS, Hidden Markov Model	Pomegranate	Bacterial blight	75.00%	Further work can be done in implementing ML algorithm for pre-processing and enhancement
Microsoft (2022)	Raspberry Pi3, Thingspeak, PSO optimized CNN	800 citrus	Canker, scab, black spot, greening, anthracnose, and melanose	96.08%	Local exploration can be avoided with other meta heuristics optimization algorithms
Khattab et al. (2019)	Sensors, ATmega2560 microcontroller having Arduino bootloader, SIM900 GSM transceiver, Grounded Layered Architecture with IR agent for forecasting	100(Toma-toes and potatoes)	Late blight, early blight, powdery mildew	0–5 (Disease Severity Index)	Research can be further investigated in automated fertilizer and precision irrigation application
Abdulridha et al. (2019)	Pika L 2.4 camera with SpectrononPro control software, DJI Matrice 600 multi-rotor UAV equipped with Pika L 2.4 camera, two classifiers (KNN and RBF)	160 citrus tissues	Growth stage of citrus canker (late, early and asymptomatic)	92.00%	Vegetable indices can be further explored to determine the other harmful diseases

(*continued on next page*)

Table 8 (*continued*).

Pavel et al. (2019)	Three sensors with camera, Arduino Mega 8560 microcontroller, Raspberry Pi 3, Multiclass SVM for classification	150 (cucumber, tomato and eggplant)	Anthracnose, bacterial blight, alternaria, dislocation rosae, and cercospora spot	97.33%	The classifer models fail to achieve the reasonable accuracy for large datasets. Deep Belief network led to future work
Xiao et al. (2022)	UAV (DJI Matrice 200 multi-rotor) equipped with the MicaSense RedEdge-M multispectral camera and RF, SVM and decision tree classifiers	165 apple trees	Apple fire blight disease	94.00%	Model can be slow down for multi-disease classification
Schoofs et al. (2020)	UAV (an Altura X8 octocopter) incorporated with COSI hyperspectral camera and fixed-wing SenseFly eBee platform with S.O.D.A camera, tree-based model	47 pear orchards	Pear fire blight disease	85.00%	Further improvements can be done by focusing operational tracking of fire blight with UAV-equipped spectral sensors
Moriya et al. (2021)	Rikola hyperspectral camera integrated with UAV-UX4 model, UV/NIR ASD spectrometer (for radiometric correction), SAM classifier	7127 citrus trees	Citrus gummosis disease caused by fungus Phytophthora spp	Multispectral: 79.00% (Accuracy), 0.55 (F-Score) Hyperspectral: 94.00% (Accuracy), 0.85 (F-Score)	The accuracy of the Multispectral data can be further improved by choosing specific bands that are more susceptible to citrus gummosis disease
Devi et al. (2020)	IOT sensors (Temperature, pH, Gas and Moisture), Arduino UNO as a microcontroller, ThinkSpeak cloud server and GoogLeNet architecture	Multiple fruitage varieties	Determination of the pesticide amount in various fruits & vegetables	99.90%	Generalization of the model in detecting of other harmful diseases with same accuracy can lead to the base of the future work
Dairath et al. (2023)	Four channel relay modules (LED, Roller pump, Gear motor, Roller pump), four servo motors as a robotic arm	Multiple species	Binary classification (Good or Bad)	15.00 s (Detection time)	Improvise the model by implementing roller conveying module to visualize the 3D view of species
Vanitha et al. (2023)	Color frequency IR sensor, conveyer moving belt, servo motor, LED, Arduino UNO(IDE), Rectifiers	Tomatoes	Ripe or unripe	20.00 s (mean count time)	Model can be transformed with more parameters like recognizing the defective & disease type
Zhu et al. (2023)	Knowledge graph, Raspberry Pi camera module, Neo4j storage, Hanlp toolkit, VGG16 classifier	Lychee & Longan	Sooty blotch disease entity, hypitima longanae pest entity, & colletotrichum spp bacteria entity	94.90%	Large parameter size of the pre-trained model can be further replaced with other CNN models to extend the accuracy

were fused using the cross-fusing method to construct DMCNN. Mishra et al. (2019) analyzed the heat index value to differentiate the ripened stage of lime, lemon, guava, orange, plum, green peas, potato, carrot, cucumber, and carrot. The IoT architecture was developed using an optical sensor integrated with an infrared camera for real-time infrared image acquisition, an Arduino processor for heat index, and a Wi-Fi module for communication over the network. The ripe, unripe, and rotten species were investigated based on the heat signature parameter. The validation of the proposed architecture can be further tested against multiple large-scale varieties of fruits and vegetables. Also, further research can be carried out to emphasize additional parameters of fruits other than the heat index.

Furthermore, the YOLOV3 object detection model for autonomous maturity level detection of the strawberry was analyzed by Zhou et al. (2021). The proposed method collected three datasets from UAV at 2 and 3 meters height using DJI Phantom 4 Pro equipped with RGB cameras that were further normalized using an adjusted reflectance instrument. Similarly, near-ground digital images were captured using a Canon EOS 5D Mark III camera at a height of 0.5 m. Thereafter, a pre-trained YOLOV3 model was used to detect the seven different maturity stages (from flower to rotten) for RGB images, whereas, three different maturity stages (flower, mature, immature) were analyzed for the UAV images. Akhter and Sofi (2021) incorporated IoT-based wireless sensor networks to identify apple scab disease using ML algorithms. Real-time data acquisition was performed using six DHT22 IoT sensors (S_1 to S_6) with an MIB 520 embedded IRIS 2.4 GHz module and Zigbee protocol to record temperature and humidity. Likewise, Jawade et al. (2020) introduced a random forest and IoT-based model for early disease prediction in Alphonso mangoes using ATMEGA microcontroller. Mango disease prediction was made by experimenting with a random forest classifier against real and past data of 20 years from the meteorology office of Vengurla, Sindhudurga.

Moreover, IoT-based prototypes can be used for fruit sorting purposes. Ismail and Malik (2022) developed a prototype consisting of two Raspberry Pi components for inspecting the grading of apples and bananas based on the disease severity levels. The training and testing images were segmented using shift and watershed methods respectively (Cheng, 1995; Vincent and Soille, 1991). The classification performance was tested against five state-of-the-art models, namely, EfficientNet, MobileNetV2, ResNet, NASNet, and DenseNet. The experimental results illustrated that the EfficientNet model outperformed four other methods by achieving an accuracy of 93.80% and 96.70% for bananas and apples, respectively. Furthermore, Arya and Gangwar (2021) investigated the ripeness severity of the apple, banana, and orange. Different real-time sensors were used to collect the parameters like pH level, temperature, time, and ripeness.

On the same footprint, a novel Raspberry Pi-3-enabled Compressed Sensing Web-Enabled Disease Detection System (CS-WEDDS) was addressed for the disease detection problem (Aasha Nandhini et al., 2018). The diseased images were segmented using a Mean-based Threshold Strategy (MTS), which was further processed using the DCT algorithm to generate sparse matrices (Indumathi et al., 2017). Further, the storage overhead was mitigated by employing the hybrid measurement matrix method (Nandhini et al., 2015). Finally, texture features were extracted using the GLCM, which was passed to a linear kernel-based SVM classifier. Likewise, a smart IoT system was designed to capture images using an RGB and thermal camera-SEEK with a thermal sensor (Bhole et al., 2020). During data acquisition, 11 different agro-products were revolved on the tray fixed at the top of a stepper motor to capture 3D video frames. The stepper motor was controlled by an

Arduino Uno-microcontroller attached to ICA4988. Eleven outer texture features were collected using GLCM, which were further classified into infected and non-infected classes by random forest and KNN. The experiments concluded that the random forest classifier overshadowed the KNN algorithm for texture features of both RGB and thermal images.

Gawas et al. (2021) suggested an intelligent farming system for monitoring the freshness level of apples, bananas, and oranges. The architecture consisted of a conveyor belt that was integrated with a motor, sensors, three cameras, and nodeMCU8266. MQ3 and MQ4 sensors determined the species's ethanol (C_2H_6O) and methane gas (CH_4) concentrations. Transfer learning with the inception V3 model was leveraged for recognition and disease classification. Similarly, An IoT-cloud-based method was developed for early disease detection in pomegranate tissues (Pawara et al., 2018). Digital temperature sensor (DHT22) and humidity sensor (AM2302) were used to collect the real-time temperature and humidity values, which were processed by Arduino Uno microcontroller. The processed images were sent to the ThinkSpeak cloud. The framework then fed the captured cloud dataset to the Hidden Markov Model (HMM) using a base station to predict the bacterial blight disease based on probability distribution.

Recently, swarm-optimized IoT models have received significant attention from researchers for the parameter optimization of IoT-based smart systems in agriculture. Microsoft (2022) implemented a PSO-optimized CNN with IoT modules for multiple disease detection in citrus tissues. Real-time citrus images were collected using camera sensors and transferred to the ThinkSpeak cloud via the Raspberry Pi3 IoT interface. The Adam-optimized cross-entropy loss function was evaluated as a fitness function for the CNN network. During the velocity computation phase, the variation of pooling layer parameters from the fully connected layer is determined by decision factor C_g. If $r \leq C_g$ then variation from the pooling layer (gBest) to the current layer is chosen; otherwise, the variation from the fully connected layer (pBest) to the current is chosen where $r \in [0,1]$. If the current layer parameters have comparable layer types to gBest and pBest, the velocity computation is applied. The developed method outperformed other considered methods by achieving an accuracy of 96.08%.

An agro-weather system was developed for automatic disease detection of potato and tomato vegetables (Khattab et al., 2019). The proposed system acquired eleven real-time attributes relevant to the tomato vegetables. ATmega2560 microchip integrated with an open-source Arduino bootloader was used to send the sensor's information (ATmega, 2022). The back-end layer implemented a software-based expert system for disease prediction. The software module exploited the GLAIR AI agent (Grounded Layered Architecture with Integrated Reasoning) for forecasting the results (Shapiro and Bona, 2010). Abdulridha et al. (2019) utilized ML-enabled smart UAVs for estimating late, early, and asymptomatic symptoms of canker disease in citrus tissues. Indoor hyperspectral images were collected using an imaging setup consisting of a two-dimensional Pika-L-2.4 camera and SpectrononPro control software. Similarly, Outdoors hyperspectral images were collected using a two-dimensional Pika-L-2.4 camera and UAV with DJI Matrice 200 multi-rotor and Pro Hexacopter. The captured hyperspectral images were calibrated to radiance images using an Imager Calibration Pack (ICP) installed with Spectronon Radiance Conversion (SRC) plugins.

Pavel et al. (2019) utilized IoT-based sensors for real-time image acquisition of tomato, cucumber, and eggplant tissues using the camera interfaced with the Arduino Mega 8560 microcontroller. The resized RGB images were further transformed to the L∗a∗b color subspace. Finally, the images were segmented to identify the disease-infected regions using the K-means clustering algorithm. Xiao et al. (2022) introduced a smart UAV with a DJI-Matrice 200 multi-rotor and the MicaSense RedEdge-M camera for multispectral image acquisition. The images were calibrated using Pix4DMapper software for radiometric and geometric calibration to generate the orthophotos and digital surface model. The top three Vegetation Index (VI) features, namely, Triangular Vegetation Index (TVI), Anthocyanin Reflectance Index (ARI), and Ratio Vegetation Index (RVI), and their optimal correlation were determined using the minimum Redundancy Maximum Relevance (mRMR) method. Additionally, iForest (isolation Forest) algorithm was used to identify the fire blight disease in the apple (Liu et al., 2008).

On the same footprints, a SAM and GNSS-based model was introduced by Moriya et al. (2021). The hyperspectral images with twenty-five sensor bands and multispectral images with three sensor bands were acquired from Rikola DT-0014 camera integrated with the UAV-UX4 model (HSI, 2022) to generate hyperspectral cubes. The radiometric calibration and ortho-calibration of hyperspectral cubes were performed by UV/NIR ASD spectrometer (Honkavaara et al., 2013). The processed images were passed to investigate the health level of the citrus trees. Furthermore, Devi et al. (2020) delineated the IoT system in which real-time fruitage parameters (MQ135 sensor for atmospheric gas, DHT11 sensor for temperature and humidity, pH sensor) were collected, which were pre-processed using Arduino Uno microcontroller. The processed features were saved on the ThinkSpeak cloud via the ESP8266 wireless communication module. The result showed that GoogLeNet architecture performed superior as compared to the other considered methods.

A novel robot prototype was introduced to pluck agro-products based on disease severity level (Dairath et al., 2023). However, the tissue defect percentage was calculated to select the threshold value. The grading result was further transmitted to the robotic arm via Arduino Uno microcontroller. Moreover, autonomous defect detection of tomatoes was categorized into ripe(red) and unripe(green) based on the color vision and IoT system (Vanitha et al., 2023). The servo motor moves the tomato to the conveyor's right side. The IR Sensor counted the number of ripe and unripe tomatoes, whereas the color was decided based on the frequency of the radiation. A recent IoT-based system was developed for the pest identification of longans and lychees using Raspberry Pi and VGG16 network (Zhu et al., 2023). The important features were extracted using the Hanlp toolkit, whereas the sooty blotch as a disease entity, hypitima longanae as a pest entity, and colletotrichum spp as a bacteria entity were used to develop the knowledge graph. The cosine similarity score was used for knowledge fusion which was further stored using the Neo4j database.

7. Data sets

Fruits and vegetables are prone to disease as human beings. Accurate and rapid fruitage disease categorization and early disease recognition as per their physical appearance (color, shape, and size), index, and quality parameters are the unfulfilled demands of Agriculture 4.0. With life-long shifts in the environmental scenarios, ecologically adaptable agro-product cultivars are becoming crucial in agriculture. Thus, it is necessary to assess the features at every growth phase to select potential species that can be utilized in recent technologies for multiple-dimensional areas. This section aims to develop the metadata of existing fruit and vegetable datasets with considerable disease categories, which may help the domain researchers to get consolidated one-place information for the study. Table 9 presents some of the names, properties, acquisition methods, and URLs of the benchmark fruitage dataset.

7.1. Fruit and vegetable datasets

The Guava tissue dataset was prepared by Rajbongshi et al. (2022) for real-time disease classification. The dataset contained resized images having four categories of fruit (Healthy, Scab, Phytophthora, and Styler end rot) and two classes of leaves (Healthy and Red rust). The dataset consists of 681 images divided into five disease classes, namely, Scab (106), Phytophthora (114), Styler end rot (94), Red rust (87),

Table 9
Fruit & Vegetable benchmark datasets.

Researcher	Species or Datasets	Total	Dataset properties	Dataset collection	Dataset URL
Rajbongshi et al. (2022)	Guava fruits and leaves	681	114 Phytophthora, 94 Styler end rot, 106 Scab, 87 Red rust, 126 Healthy leaves and 154 Healthy fruits	Nikon DSLR D3200 camera with 1.5 time focal length and 23.2 × 15.4 mm resolution	https://data.mendeley.com/datasets/x84p2g3k6z/1
Rauf et al. (2019)	Citrus fruit and leaves	759	15 Scab, 241 Canker, 220 Greening, 190 Black spot, 13 Melanose and 80 Healthy images	Canon DSLR EOS 1300D camera with 14.9 × 22.3 mm sensor size and 5202 × 3465 resolution	https://data.mendeley.com/datasets/3f83gxmv57/2
Meshram and Patil (2022)	Multiple fruits varieties	19 526	High-quality images of six categories having three variants i.e. Good, Bad and Mixed quality	High-resolution iPhone6(Apple), Realme(5 Pro) and ZUK(Z2 Plus)	https://data.mendeley.com/datasets/b6fftwbr2v/1
Kumar et al. (2021)	Pomegranate	90	Three-grade value of three different quality of fruit individually.	Logitech C905 720 pixel webcam with 2 MP sensor having 1600 × 1200 resolution	https://www.kaggle.com/kumararun37/pomegranate-fruit-dataset
Medhi and Deb (2022)	Multiple categories of Banana	8000+	2343 leaf images, 102 Stem images and 216 fruit images for different variants of banana	Samsung SM-G610F with 9.6 MP and Nikon SX 70 with 18.3 MP	https://data.mendeley.com/datasets/4wyymrcpyz/1
Sara et al. (2022)	Cauliflower	656	177 Downy mildew, 173 Bacterial spot rot, and 100 Black rot, and 206 Healthy images	Sony Cyber-Shot-W-530 model with 14 MP and 500 × 500 pixels resolution	https://data.mendeley.com/datasets/t5sssfgn2v/3.
Tariq et al. (2022)	Fruitfly species	2000	Dataset contains 2000 images of Bactocera dorsalis and Bactocera zonata	Oppo F11 smartphone with 48 MP camera and Raspberry Pi camera with 8 MP	http://dx.doi.org/10.17632/hgz2n5jxhp.1
Kaufmane et al. (2022)	Japanese quince	1515	Raw Japanese quince of 650 ripe and 889 unripe images	Samsung Galaxy A8 mobile phone having 3456 × 3456 pixels resolution	https://zenodo.org/record/6402251
Alencastre-Miranda et al. (2018)	Sugarcane	786	six class like Two buds(green), One bud(orange), No buds(yellow), Cracked(red), Crushed(blue) and No damage(no marking)	Stereovision camera (ZED from Stereo Labs), CCD camera (Blackfly S from Point Grey Research) & a scanning laser rangefinder (Hokuyo)	https://github.com/The77Lab/SugarcaneBilletsDataset
Kaggle (2022)	Apple, Orange, and Banana	13 599	2088 (fresh apple), 1962(fresh banana), 1854(fresh orange), 2943 (rotten apple), 2754 (rotten banana), 1998 (rotten orange)	Public dataset	https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification
Fenu and Malloci (2021)	Pear	3505	3006 leaves images and 499 fruit images with different severity level	Canon DSLR EOS 60D camera with 3456 × 5184 pixels resolution and Honor 6X mobile with 2976 × 3968 pixels resolution	https://zenodo.org/record/5557313#.YtEgQ3ZBy5d
Das et al. (2020)	Tomato	6470	6470 tomato images of 10 classes. Tomatoes are graded from 1 to 10 class where 1 represents fresh and 10 represents rotten	Public dataset	https://github.com/skarifahmed/FGrade
Rauf and Lali (2021)	Guava fruits and Leaves	306	70 Rust, 77 Canker, 83 Mummification, and 76 Dot guava images	Images are acquired with the mobile phone having 300 dpi resolution and 6000 × 4000 size	https://data.mendeley.com/datasets/s8x6jn5cvr/1
Munasingha et al. (2019)	Papaya	214	55 Anthracnose, 24 Black spot, 43 Phytophthora, 42 Powdery mildew, and 50 Ringspot	Public dataset	https://data.mendeley.com/datasets/7dxg9n2t6w/1
Mendeley (2022)	Eight types of fruits	3200	3200 fruit images of fresh and rotten category	Public dataset	https://data.mendeley.com/datasets/bdd69gyhv8/1
Meshram et al. (2020)	Six types of fruit	12 000	1000 images of each fruit (Good/Bad Quality) having 256 × 256 size adding up 12 000 images	Public dataset	https://ieee-dataport.org/open-access/fruitsgb-top-indian-fruits-quality
Altaheri et al. (2019)	Date	8231	First dataset contains 8079 images from immature to mature states. The second dataset contains 152 Barhi date packs belonging to 13 palms	Three high-quality cameras: Canon EOS-1100D, Canon EOS-600D and Sony HDR-CX405 with different illuminance conditions.	https://ieee-dataport.org/open-access/date-fruit-dataset-automated-harvesting-and-visual-yield-estimation
Oltean (2021)	Multiple varieties of fruits	90 483	67 692 images and 22 688 images for training and testing purposes	Image acquisition was performed using Logitech C920 camera having 3 rpm shaft speed and 100 × 100 pixels resolution.	https://data.mendeley.com/datasets/rp73yg93n8/1 https://www.kaggle.com/datasets/moltean/fruits

(continued on next page)

Table 9 (*continued*).

Gené-Mola et al. (2021)	Apple	640	615 real-time Fuji apples and 25 more apple images under laboratory conditions.	Canon EOS 60 DSLR camera having 18 MP CMOS APS-C sensor	https://dataverse.csuc.cat/dataset.xhtml?persistentId=doi:10.34810/data141
Tian et al. (2021)	Apple	5801	Total of 2720 healthy apple fruits and 3081 images infected from various diseases	Public Dataset	https://gitee.com/cheng_xiao_yuan/AI-Challenger-Plant-Disease-Recognition
Mamdouh and Khattab (2021)	Olive Fruit Fly	1680	DIRT dataset consists of 848 olive fruit fly images and Leeds contains 832 butterfly images in PNG Format	DIRT Olive Fruit Fly DataSet and Leeds butterfly dataset	http://lefkimi.ionio.gr/∼avlon/dirt-dacus-image-recognition-toolkit/ http://www.josiahwang.com/dataset/leedsbutterfly/
Li et al. (2022)	Jute	4418	3862 images and 556 images for training and testing purposes	Central Research Institute for Jute and Allied Fibres	https://crijaf.icar.gov.in/
Vasumathi and Kamarasan (2021)	Pomegranate	6519	2813 Healthy and 3706 Unhealthy Pomegranate images	iPhone and DSLR Camera	http://www.cofilab.com/portfolio/image-database-pomegranate https://www.kaggle.com/datasets/kumararun37/pomegranate-fruit-dataset
Han et al. (2021)	Apple	74	Each image contains a bunch of apples provided at least one apple must be infected with Anthracnose disease.	The dimension of disease symptoms varies from 50 × 50 to 2000 × 1400, and the dimension of apple images varies from 500 × 700 to 3000 × 2000.	https://github.com/QuIIL/Dataset-Region-Aggregated-Attention-CNN-for-Disease-Detection-in-Fruit-Images/tree/main/Images
IFW (2022)	Apple fruit and Leaves	8632	5244 Infected and 3388 Uninfected apple images	CASC-IFW datasets	https://engineering.purdue.edu/RVL/Database/IFW/index.html

and Healthy (280). Rauf et al. (2019) introduced a novel citrus plant dataset for fruit, stem, and leaves. The researcher collected 759 images of citrus tissues (fruits and leaves) following Healthy (22 & 58), Canker (78 & 163), Black spot (19 & 171), Greening (16 & 204), Scab infected fruits (15), and Melanose infected leaves (13). The FruitNet dataset consisting of 19 526 images was developed by Meshram and Patil (2022). The dataset contained images of six different species, namely, banana, lime, apple, orange, pomegranate, and guava. The images were categorized into Good, Bad, and Mixed classes, with 11 664, 6778, and 1074 images, respectively.

Further, grading is a similar term used to inspect the quality of fruits and vegetables. Kumar et al. (2021) created a pomegranate dataset of ruby cultivars. The dataset was divided in three gradings (G1, G2, & G3) and four qualities classes (Q1, Q2, Q3, and Q4). The grades were decided based on the tissue's color, size, weight, and physical appearance, where G1, G2, and G3 signified the superior and good appearance with complete ripeness. The database comprised 30 sample images from each grade making up 90 pomegranate images. Das et al. (2020) introduced a tomato grading dataset. A total of 6470 tomato images were collected for 90 days and graded from labels 1 to 10. The classification accuracy of the benchmark dataset was also compared against different state-of-the-art models.

Some datasets cover the different parts or cultivars of plants infected by the pathogens instead of the edible part of the fruit and vegetable. PFSD (Medhi and Deb, 2022) dataset consisted of different banana parts (stems, leaves, flowers, and fruits). The images infected from Scarring beetle (150), Panama disease (102), Pseudo-Stem Weevil (2736), Bacterial soft rot (1078), Yellow Sigatoka (264), Aphids (366), and Black Sigatoka (474) were acquired under natural light. Moreover, 102 stem images, 2343 leaf images, and 216 fruit images of four banana varieties (Jahaji, Bhimkol, Malbhog, and Kachkol) were collected. Moreover, Altaheri et al. (2019) acquired a date palm dataset using three cameras under natural sunlight from June to September for yield detection, disease classification, and robotic farming. The first dataset covered 8079 images belonging to different variants (Sullaj, Khalas, Meneifi, Barhi, and Naboot Saif) with varying stages of maturity (Tamar, Rutab, Khalal, and Immature). The second dataset contained 152 Barhi dates with 13 palms at different ages and heights.

VegNet cauliflower dataset was introduced by Sara et al. (2022). The dataset was divided into four classes, namely, Downy mildew (177), Bacterial spot rot (173), Black rot (100), and Healthy (206), which was increased to 1770, 1730, 1800, and 2060 respectively, by applying the data augmentation method. Similarly, Tariq et al. (2022) developed a fruit fly dataset containing 2000 images of Bactrocera dorsalis, and Zonato species. The resized images of 416 × 416 pixels were collected through Raspberry Pi cameras equipped with an intelligent trap tool. The YOLOV5 model was used for species detection and annotation with a mean accuracy of 85.00%. Kaufmane et al. (2022) designed the "QuinceSet" Japanese quinces dataset containing 1515 RGB images that were labeled as class 0 (ripe) and class 1 (unripe) for phenotyping and detection. The dataset contained raw images derived from ripe (650) and unripe (889), which were annotated to 17 171 using state-of-the-art YOLO architecture.

The quality and productivity of the fruitage crops are also affected due to physiological disorders. The sugarcane billets dataset was developed by Alencastre-Miranda et al. (2018). The dataset consisted of 120 sample images, categorized into six classes, namely Healthy, Cracked, Crushed, Two buds, One bud, and No bud. The labeling was performed based on no visual marking, red marking, blue marking, green marking, orange marking, and yellow marking, respectively. DiaMOS (Fenu and Malloci, 2021) dataset contains 3505 pear images (3006 leaves and 499 fruits) from Septoria Piricola species. The images were collected during the entire planting season from February to July 2022. The fruit images were categorized into four classes: ripening, nut fruit, fruit set, and fruit growth. At the same time, leaves were classified into Healthy, Unhealthy, Spot, and Curl classes. The image annotation was carried out through the YOLO model.

Rauf and Lali (2021) developed an image gallery of 306 guava images. The dataset was classified into four disease class labels, namely Rust (70), Canker (77), Dot (76), and Mummification (83). Furthermore, a digital library of papaya tissue datasets containing 214 images was developed by Munasingha et al. (2019). The dataset was augmented into five defect classes: Black spot, Powdery mildew, Phytophthora, Ringspot, and Anthracnose disease. The CNN model was employed for papaya disease multi-classification with nearly 92.00%, which outperformed the SVM architecture. The powdery mildew class took the maximum time for classification, whereas the black spot disease recorded the minimum time.

Similarly,multi-fruit datasets were also developed to build a robust disease classification system. Kaggle (2022) created a banana, apple, and orange dataset of 13 599 images for binary classification (rotten or fresh). The dataset can be summarized as fresh images (2088 apple, 1962 banana, and 1854 orange) and rotten images (2943 apple, 2754 banana, and 1998 orange). Mendeley (2022) designed an extensive database with eight varieties of fruits (apple, orange, strawberry, banana, guava, grape, pomegranate, and jujube). The dataset was labeled into fresh and rotten classes for binary classification. The dataset had 3200 original images, which were increased to 12 335 images with basic augmentation methods such as rotate, scale, translate, and blurring. Meshram et al. (2020) generated the "FruitGB" dataset of six popular agro-products: apple, orange, lime, banana, guava, and pomegranate. The dataset contained 6000 multi-fruit images divided into good and bad classes.

Han et al. (2021) collected 74 apple images to analyze the binary fruit defect classification (Anthracnose or Non-anthracnose) with an improved Mask R-CNN network. The images were annotated using 1 to 9 rounded boundary regions per image. IFW (2022) developed a CASC-IFW database of 8632 for autonomous disease detection in multiple variants of apple fruits (Golden Delicious, Red Delicious, York, and Fuji). Oltean (2021) processed the images using a flood fill algorithm for background noise removal. The proposed dataset can be used for multiple dimensions: ripeness determination, clustering, defects, edible (binary classification), and maturity detection, seed quantity, growth stage, hardness level, and fruitage type (multiclass classification). A total of 131 classes from different fruits, seeds, and vegetables, summing up to 90 483 images, were incorporated into the database. Gené-Mola et al. (2021) developed PFuji images with 3D data points during the growth phase and annotated them. The dataset was designed using 303 apple trees (2018) and 312 apple trees (2020), thus making up 615 real-time images. The dataset was divided into four phases first phase (raw images), the second phase (data points from the east and west side), the third phase (merging and segmentation of 3D points), and the fourth phase (Apple tree information and location).

Similarly, the apple dataset was developed by Tian et al. (2021). The dataset comprised leaf images (Healthy (1354), General scab (1205), Severe scab (1026), General cedar (972), Serious cedar (460), and Spot (488)), and fruit images (Healthy green (1360), Healthy red (1360), Ring rot (93), General anthracnose (78), and Severe anthracnose (87)). Mamdouh and Khattab (2021) designed an olive fruit fly species detection framework utilizing YOLOV4 for object detection and localization with a mean accuracy of 96.68%. The DIRT image dataset with 848 images (positive samples) was collected from liquid aromatic McPhail traps. The Leeds butterfly image dataset containing 832 images (negative samples) of Danaus plexippus species was acquired from the internet.

A jute dataset infected for multiple diseases and pest detection using YOLO-JD architecture was introduced by Li et al. (2022). A total of six fungal diseases (Black band, Dieback, Tip blight, Soft rot, Anthracnose, and Stem rot), two viral diseases (Chlorosis and Mosaic), and two pests (Comophila sabulifers and Hairy caterpillar) were appended to the database, which was pre-processed (filtering and illumination correction) for better accuracy. Vasumathi and Kamarasan (2021) created a novel pomegranate dataset that combined real-time and existing datasets for binary disease classification (normal and abnormal) using deep CNN and LSTM models. The dataset combined 328 images from the Cofilab dataset and 90 images from the Kaggle dataset, while the rest of 6101 pomegranate images were acquired under real-time environmental conditions.

8. Research gaps, challenges, and future opportunities

The survey comprehends the significance of smart techniques to identify complex problems in the fruitage disease identification domain. The state-of-the-art algorithms have shown remarkable outputs by minimizing produce losses in farming. Agriculture productivity is enormously increasing while utilizing such techniques, but the benefits come with its cost. A few of such challenges with their viable solutions are mentioned below:

8.1. Machine learning research gaps and future directions in fruitage domain

- **Data Pre-processing**: Fruitage data is one of the key components for the success of any machine model. Several public digital repositories for fruits and vegetables are available for the use of researchers. Generally, the real field fruitage data is acquired using UAV, satellite, or mobile camera in different seasons. Thus, these fruitage datasets are present with several limitations, such as improper data formatting, poor quality, incompatibility, imbalanced or noisy data, redundant features, hazed, low resolution, and unknown data values. Thus, exquisite and robust pre-processing techniques are required to incorporate the eligible data for testing, validation, and training of the models, which is time-consuming and error-prone. The literature witnessed the challenges of fruitage data pre-processing, like spatial loss information, color distortion, and over-enhancement for disease classification model (Habib et al., 2020). The majority of research work studied the impact of extracting the best statistical and texture feature maps of the dataset for better classification (Bhargava and Bansal, 2020a; Mojumdar and Chakraborty, 2021; Almadhor et al., 2021). Different methods like Gaussian filtering, color histogram equalization, segmentation, and color space conversion help to perform the data preprocessing for precise fruitage defect detection model convergence (El-aziz et al., 2020; Roy et al., 2021b; Ghodke et al., 2022; Bhargava and Bansal, 2020a). Such techniques can also exaggerate the background noise, giving the fruitage image a noisy appearance in the region of interest. Also, multi-modal data is another room for improving the statistical and data interpretation methods to understand the food domain better such as fruitage waste production, crop harvest prediction, and fruit nutrient determination. The fruitage images acquired in extreme weather conditions such as rain, fog, and night is another major challenge that may affect the model's performance. To mitigate this, image dehazing and thermal imaging methods may be introduced as effective data pre-processing techniques (Yu et al., 2018). Sequential Pre-processing Through Orthogonalization (SPORT) method may be future direction for effective data-processing that hybridizes the multiple pre-processing methods to ensure high disease classification accuracy (Mishra et al., 2020).
- **Bias & Dynamic Problems**: Generally, ML models trained on fruitage datasets suffer from the label bias problem as the majority of the real field datasets are imbalanced. Thus, the testing performance of the models does not meet the standards of the training performance. The improper selection of training and testing data samples may lead to model instability resulting in poor performance of the model in real-time scenarios. The research studies witnessed that the feature engineering methods are useful to reduce bias by incorporating the relevant and important features and removing irrelevant or discriminatory features (Bhargava and Bansal, 2020b; Kumari et al., 2021). Careful feature engineering can greatly enhance a model's ability for generalization and prediction accuracy (Shafiq et al., 2020b,c). The fruit and vegetable feature maps can be extracted using the delta E (ΔE) method (Cugmas and Štruc, 2020; Almadhor et al., 2021). The study extracted homogeneity, energy, contrast, and correlation discriminative texture features using GLCM while other relevant fruitage factors like area, perimeter, eccentricity, diameter, major and minor axis, and orientation were extracted during the feature engineering process (Dubey and Rocha, 2023; Sugiarti et al., 2021). Similarly, meteorological and ecological factors like environmental temperature and soil moisture are dynamic, making it difficult to provide an accurate prediction. The meteorological & ecological features affecting the fruitage growth like wind speed, relative humidity, solar radiation, and temperature, and the physicochemical features such as firmness, weight,

soluble solids, sugar, moisture content, and tannin content may be extracted for disease detection (Mohammed et al., 2023). The literature survey deduced that disease categories present in the fruitage domain generally suffer from maturity bias, seasonal bias, and label bias. Such data bias-ness problems may be handled using continual or incremental machine learning methods (Li and Chao, 2020). Additionally, fair machine learning algorithms for re-sampling or re-weighting feature maps, demographic parity, or equal opportunity algorithms can be employed for bias mitigation. A few statistical techniques, such as KL divergence or Kolmogorov–Smirnov test and distribution or domain adaptation can be introduced to detect shifts in data distribution that help to handle the dynamic issues in the fruitage disease detection (Hien and Gillis, 2021). Semi-supervised machine learning methods like Expectation–Maximization (EM), Self-Training, Co-Training, and Multi-View Learning leverage the unlabeled data to improve model performance (Zhu, 2005).

- **Limited Fruitage Data**: Machine learning models' training and validation process requires massive fruit data to assess models' performance before deploying in a real-world environment. Moreover, the effective training process requires high-performance hardware resources, large storage devices, and testing tools. Under-training and over-training are the open issues in the fruit industry that must be addressed while training the model. For instance, the unripe banana shows different color features than the ripe banana. Limited model generalization may result from a dataset lacking these variances, which can be controlled using regularization methods (Wang and Ni, 2019). Regularization techniques (L1 & L2), $k-$ fold validation, early stopping, decision tree pruning, and ensemble methods can be exploited to address such issues. It is also recorded from the literature that the availability of a robust fruitage dataset is a major challenge for the effective training of the model. To overcome this, traditional image augmentation methods may be used to handle the complexity of model overfitting and bias-ness towards the specific class. Rotation, flipping, shearing, and random erasing are the few instances to expand the fruitage images. Synthetic data generation methods like Synthetic Minority Over-sampling Technique (SMOTE) and Deep Synthetic Minority Oversampling Technique (DeepSMOTE) can also be used to handle class imbalance issues (Alzubaidi et al., 2023). However, recent machine learning methods such as self-supervised learning, reinforcement learning, and active learning open up the likelihood of handling smaller labeled datasets. Collaborative learning, time-lapse image acquisition, cross-domain adaptation, and multi-task learning are the open solutions to be addressed in fruit and vegetable disease detection.
- **Optimal Model Deployment**: A large number of ML models are available for classification and prediction tasks. Thus, selecting an optimal model for the specific fruit or vegetable is still challenging. The naïve approach is to select a model randomly with different attributes using a trial and error approach, which takes a long time to conclude the result. The literature survey revealed that different ML models show varied performance with different problem domains like disease classification, grading, and disease segmentation in the fruitage dataset. The lightweight classifiers such as Cubic SVM, Fine KNN, Bagged Tree, Complex Tree, and Boosted Tree can be employed for fruitage disease identification which can be trained on less number of image samples with reasonable performance (Almadhor et al., 2021). Similarly, different SVM variants like cubic-SVM (C-SVM), quadratic-SVM (Q-SVM), linear-SVM (L-SVM), and radial basis function-SVM (RBF-SVM), decision tree, ANN, and RF may be deployed for fruitage grading purpose (Ireri et al., 2019). The study utilized a self-clustered method by using the likelihood of nearest neighbor and extended Gaussian kernel approach to identify the lesions in the real-field farms (Hasan et al., 2023). The fruitage disease classification requires the segmenting of every fruit or vegetable, classifying the pixel-by-pixel disease severity, and then predicting the disease category. It is an inherently difficult task to segment individual real-time fruits due to their overlapping or occlusion. Different segmentation methods such as Histogram, random walker, entropy-based segmentation, and GMM can be applied for rotten area identification in fruits and vegetables (Roy et al., 2021b). Whereas, Watershed Transform, Fuzzy C-means clustering, and Conditional Random Fields (CRF) may be the future directions to be explored for better fruitage disease segmentation. Neural Architecture Search (NAS), a co-domain of the AutoML, automates the network's topology with limited computational resources to obtain the best disease classification network (Zhang et al., 2021). Pruning, quantization, factorization, and knowledge distillation methods may be exploited to compress the model parameters which reduces memory utilization without compromising the disease classification accuracy.
- **Model Complexity**: The computational complexity of the machine learning models is also a prime concern for agricultural applications. Linear machine learning algorithms predict the output as the linear combination of the input feature maps. They are easy to implement but do not capture the complex defect patterns in fruits and vegetables. For instance, the LR model may be exploited for disease detection in the fruits where each disease class has balanced weights or sampling and has distinct spectral or visual symptoms (Mohammed et al., 2023). On the same note, non-linear machine learning algorithms have multiple layers and can include activation functions to introduce non-linearity, but these techniques are computationally expensive with respect to hardware, software, storage devices, and data availability. An improved non-linear SVM and backpropagation model may be implemented for disease identification where high-dimensional data shows large variation towards feature diversity (Yu et al., 2018). Thus the tradeoff between the model complexity and disease classification accuracy is another room for improvement in the fruitage domain. Dimension reduction techniques like PCA, LDA, t-distributed Stochastic Neighbor Embedding (t-SNE), and Independent Component Analysis (ICA) can be utilized to minimize the number of feature maps while retaining model variance. Also, analyzing the multi-class disease problem into multiple instances of binary disease classification problems can help to ease the network's decision-making process. Evolutionary algorithms such as Ant Colony Optimization, GA, Artificial Bee Colony Algorithm, and Hybrid and Multi-objective Evolutionary Algorithms are another dimension to optimize the network parameters (Drugan, 2019).
- **Ecological Fruitage Diversity**: Increasing demand for high-quality fruits & vegetables requires a better understanding of pathogens that are indirectly affected by ecological factors. The fundamental knowledge of the ecological process is essential for the development cycle of the fruit species. For instance, climate conditions not only impact the physiology and phenology of fruits and vegetables but also affect the biotic and abiotic agents during the pollination of the crops (Tasnim et al., 2021). The ecological datasets are complex, high-dimensional, and non-linear in nature. Traditional linear statistical approaches like generic linear models are poor in identifying the complex patterns and relationships underlying ecological datasets (De'ath and Fabricius, 2000). Machine learning, on the other hand, can simulate non-linear associations in ecological data without requiring the limiting assumptions associated with parametric techniques. The survey witnessed that Classification and regression trees (CARTs) can be utilized in handling the ecological datasets (Al-Sanabani et al., 2019). Also, there are certain diseases like brown rot, blister rust, and cavity spots which are unusually available on fruits

and vegetable surfaces. One-shot and Zero-shot learning helps to develop the model to classify new disease categories while dealing with rare fruitage datasets. They can be applied to the remotely sensed data to detect ecological indicators related to the vegetation health, for instance, particular plant tissues that flourish under certain conditions. Also, they can train the model to identify the fruiting or flowering stages to track the ecological phenology.

8.2. Deep learning research gaps and future directions in fruitage domain

- **Fruitage size and Computational resources**: The fruit and vegetable data sample size is one of the prime challenges for deep learning models. For the effective training of the deep models, large training data samples are required. However, the collection of labeled fruitage datasets is a challenging task due to several dependencies such as weather, season, and pathogen attacks. Generally, the field-collected datasets are imbalanced and limited in size. As noted in the literature, the diseased classes are the minority ones in most of the repositories so it is difficult to get diseased fruitage as per the requirements in the limited tie domain. The fruitage dataset expansion may be carried out using conventional image augmentation methods, which include geometric and photometric transformation (Khalifa et al., 2022). Also, the color jittering might augment the fruit datasets by adjusting contrast, saturation, and brightness in the different color subspaces. However, these traditional augmentation methods fail to capture the feature diversity while balancing the fruitage dataset. On a similar note, Generative Adversarial Networks (GANs) have drawn remarkable attention in expanding the fruit and vegetable dataset using deep learning techniques. Vanilla GAN, Conditional GAN, Wasserstein GAN, Cycle GAN, and StarGAN are a few instances used to generate synthetic fruitage images. A novel image augmentation method known as Random Image Cropping And Patches (RICAP) may also be used to expand the training dataset from four randomly selected fruitage images (Noguchi et al., 2020). Also, the mask scoring R-CNN model can be deployed in the same direction to overcome the limited fruitage dataset problem (Yao et al., 2022). The mode collapse problem might cause the GAN neural network to generate only a subset of fruits and vegetables, disregarding the dataset's diversity. Deep learning techniques like diversity-promoting losses or progressive growing, mini-batch discrimination techniques may be used to handle mode collapse problems. The deep models are computationally complex which requires high-performance TPUs and GPUs with large storage capacity to handle enormous datasets. However, for agricultural applications, lightweight models need to be developed to meet the current demands. In the future, advanced and lightweight hybrid models may be developed to mitigate the current challenges of deep learning-based models. A lightweight CNN model having 101,000 numbers of network parameters has been developed for disease classification in banana, mango, and guava species (Hari and Singh, 2023). On the contrary, the lightweight CNN fails to capture the complex structural irregularities in the fruitage dataset. Dilated convolution is another approach to increase the receptive field without increasing the trainable and non-trainable parameters, thus helpful to develop the lightweight CNN network. ShuffleNet, SqueezeNet, EfficientNet, and MobileNets are considered as lightweight CNN models (Rangarajan et al., 2022). The adversarial noises, Progressive GANs, SRGANs, and superscaled image resolution may be deployed for better image representation.
- **Limited Fruitage Annotations**: Fruitage image annotation is an integral process during the pre-processing stage, which helps to categorize and label unknown samples using volunteers or subject-experts (which might be hard to participate). Such experts might be prone to make errors while annotating the samples, especially in identifying types of diseases in the different parts of the plants. To remedy this, the majority voting of experts may be used for labeling the images in the agricultural domain (Gené-Mola et al., 2021). However, the availability of domain experts is also one of the challenges. Thus, advanced data labeling methods may be developed in the future to annotate the data automatically. However, automated annotations may result in degenerated labeling due to missing boundary regions, redundant datasets, and outliers. The research study witnessed the image annotation by visualizing boundary box coordinates ($x_{min}, y_{min}, x_{max}, y_{max}$) to identify the severe areas in vegetables and fruits (Dhiman et al., 2022a). Deep learning methods such as YOLO, CenterNet, Faster R-CNN, and SSD (Single Shot MultiBox Detector) can be employed for real-time fruitage disease detection at multiple aspect ratios and scales. Also, image segmentation is crucial in data annotation that facilitates the disease classification process by identifying the IoU regions. It segregates the infected fruitage image into multiple segments, where each segment represents the particular pixel-level disease symptoms. Segmentation techniques like UNet, DeepLab, NDVI masking, and Mask-CNN can be exploited for multi-scale contextual information for precise pixel-level segmentation. Keypoint annotation offers detailed information about object characteristics that conventional semantic labels or bounding boxes are unable to capture (Naik et al., 2023). Comprehensive work can be done in the direction of utilizing the autoencoders and self-supervised learning methods to make efficient use of unlabeled fruit and vegetable data. Also, a novel semi-automated segmentation method may be employed to label the real-field farms to overcome the manual annotation problem (Hasan et al., 2022).
- **Hyperparameter Tuning**: Most of the DL models trained on real-field collected images require a high number of hyperparameters for effective training. The number of convolution layers and filters for a particular problem remains the pertinent issue which is solved using the naïve hit & trial method. Thus, optimal parameter selection is still an open problem, which can be ameliorated using pre-trained architectures with transfer learning and multi-layer extreme learning mechanism (Park and Yang, 2019). Several performance parameters like input bias, learning rate, and output bias, which reduced the complexity of updating the layers' weight, have shown remarkable breakthroughs for fruit disease classification (Vasumathi and Kamarasan, 2021). The transfer learning techniques might be utilized to extract optimal features to ensure multi-task learning like grading or maturity detection, fruit disease, and their severity level detection (da Costa et al., 2020). However, classifying diseases in fruits or vegetables with different backgrounds (e.g. farmland or grocery store) may further require extra fine-tuning to address such issues. Hyperparameter selection or tuning is still an iterative method that requires hit and hit-and-trial approach to finding robust models. Grid search optimizes the hyperparameters where model performance is validated using k fold validation by searching all possible permutations of the hyperparameters, whereas random search optimizes the hyperparameters using random values within the valid range of each hyperparameter. In the future, Gradient-Based and Bayesian optimization methods may be explored for determining the optimal values of the hyperparameters used in the networks.
- **Complex Fruitage Data**: Generally, the deep models are not able to produce significant results in detecting images containing occluded or tiny fruitage objects. The pooling layer of the deep models reduces the feature vector twice in each iteration, causing the model to forget the features of past layers. Thus, the deep model performance degrades as the fruitage data complexity increases, thus limiting the utilization of the pre-trained object

detection deep models in real-world circumstances. RNNs and LSTM networks are designed to handle sequential and temporal data about the fruit defects to capture data dependencies and temporal trends (Vasumathi and Kamarasan, 2021; Dhiman et al., 2023; Khattak et al., 2021). These models can utilize time series data to monitor the health and disease outbreaks in the fruitage industry. The fruitage datasets have tiny regions, which play important roles in decoding the output label. Thus, attention-based networks need to be designed in coordination with the passage blocks to remember the important features of the successive layers, which give more attention to the region of interest. The SE (Squeeze and Excitation) module might be deployed for readjusting the aggregated feature maps via the attention technique for the best feature map representation (Han et al., 2021). Further work can be carried out to design models implementing transformer models with its core self-attention and multi-head attention blocks (Rahman et al., 2018). They assign different attention weights to the various fruit defects based on the severity, thus, enabling the classification model to concentrate on important long-range dependency information. Sliding window mechanism, patching, hierarchical representations, self-attention mechanism, and positional encoding help in identifying the tiny lesions in fruits & vegetables. The complex background of the real-time fruit and vegetable images may result in the low disease recognition accuracy which can be further improved by hybridized CNN-transformer models for multi-scale local and global feature extraction.

- **Model Complexity**: The complexity of the deep learning models encompasses various factors, including the number of CNN layers, the number of neurons or activation units in each layer, and the overall architecture's width and depth. Stabilizing model complexity is crucial in deep learning. The literature survey concluded that simple deep learning models suffer from the underfitting problem, while complex deep learning models may overfit the dataset. Vegetables and fruits are vulnerable to a wide variety of diseases. Thus, complex techniques are required to control defects while lowering the use of pesticides. Building models with the right complexity requires the best model selection, dropout, early stopping, hyperparameter tuning, and regularization as per the problem domain. The convolutional neural network might be regularized with ridge loss to overcome the gradient penalty loss for better disease detection (da Costa et al., 2020). An EfficientNet deep learning model could scale the model in all dimensions by utilizing the compound coefficient (ϕ) while preserving the model complexity (Tan and Le, 2019). Pointwise convolutions having 1×1 kernel size preserve spatial dimensions while lowering the number of channels, thus minimizing computation time. Also, conditional convolutions reduce processing time by adapting the number of convolutional filters based on the properties of the fruitage dataset. Deep learning models that can efficiently incorporate information from several modalities like images, videos, and text might open up novel avenues for tasks such as multimodal learning and reasoning.
- **Model Transparency**: The intrinsic complexity of the DL models often leads to the lack of transparency about the fruits, thus making it tough to understand the decisions made by the models. The agrarians fail to understand the reasons behind the ambiguous model's prediction of a particular disease in fruits or vegetables. Thus, it is challenging to take preventive actions or confirm the disease classification accuracy without expert knowledge. It hinders the ability to identify the model bias, like overfitting to specify visual cues that might be caused by the background rather than fruit defective features. Also, the model's trustworthiness is crucial for the adoption of smart techniques in agriculture (Albahri et al., 2023). The farmers may face skepticism in automated fruitage disease classification systems due to a lack of trust in the black-box deep models. The Explainable AI (XAI) algorithms integrated with DL methods have been used to overcome these issues. Local interpretable model-agnostic explanations (LIME) (Hamilton et al., 2022), Shapley additive explanations (SHAP) (Chen et al., 2022), and t-SNE (Van der Maaten and Hinton, 2008) are few instances for the visualization of interpretable DL methods. The LIME visualization explains the predictions for each disease category by locally approximating it with an interpretable DL model, whereas SHAP visualization quantifies the importance of each lesion feature to the classification based on the Shapley values obtained from cooperative game theory. The dimension reduction technique (t-SNE) transforms the high-dimensional space into a low-dimensional space (two or three dimensions) by optimizing the K-L divergence between them.

8.3. Internet-of-things research gaps and future directions in fruitage domain

- **Data Monitoring**: IoT applications use different real-time sensors to generate huge fruitage datasets. Monitoring intelligent sensors simultaneously in arable farming is a challenging task. Significant research has been performed to implement different remote sensor systems in acquiring the disease symptoms (such as scabs, ring rot, anthracnose), physiological indicators (such as pigment and water content), and structural deformations caused by fruitage pests. It is clearly visible in the survey that fruitage disease classification is an important step for diagnosing nutrient deficiencies, projecting fruit & vegetable production, and providing a decision-making foundation for agricultural crop policy formulation. It may be addressed by deploying a multitemporal model for comparing fruits and vegetables at different phenological growth stages to measure multiple traits like pH, ripeness level, water stress, and nutrient level. Sensors onboard UAVs that utilize hyperspectral/multispectral/infrared imaging techniques may provide an excellent solution for fruitage image acquisition. Such non-destructive spectral technology may be further explored to detect multi-disease classification in the agriculture farmlands (Moriya et al., 2021). IoT device deployment indicates that the performance of the installed transceiver is overemphasized by environmental factors like wind and humidity where the wired node wants to interface. Digital temperature sensors like DHT22 and humidity sensors, namely AM2302, had been utilized to collect the in-situ information of the agriculture fields (Pawara et al., 2018). Different real-field attributes of vegetable or fruit species, namely, soil and air temperature, air moisture level, soil electric conductivity, wind direction, water level, leaf wetness, and solar radiation play crucial roles in disease detection (Khattab et al., 2019). IoT sensors (Biological air sensor DHT22 for air dampness and temperature, Gravity simple pH sensor, and Grove dampness sensor for soil dampness) can be further explored to acquire real-time fields information (Pavel et al., 2019). Various IoT components, like temperature sensors (LM35), moisture sensors (LM358), humidity sensors (DHT11), light intensity sensors (BH1750), and water flow sensors (YF-S201) are commonly used sensors to build a robust disease recognition system (Nagasubramanian et al., 2021). Also, soil NPK sensors can also to utilized to determine the real-time nutrient content of fruits. Designing energy-efficient IoT sensors by using low-energy microprocessors and power-saving modes may be considered particularly in remote areas. TinyML allows the ML models to run on low-power IoT devices without relying on powerful cloud servers and high computing hardware or software devices to address the high energy consumption issue.

- **Standardization or Interoperability**: Since billions and trillions of heterogeneous smart devices are connected to the physical world. Thus, standard communication protocols are one of the prime concerns for the researchers developing intelligent IoT frameworks for fruitage disease monitoring. Different IoT sensors might have distinct data formats, varying data modalities, and different dataset acquisition frames, resulting in misalignment and misclassification of diseases. It is crucial to synchronize the fruitage data from different sensors to produce coherent data. Thus, an appropriate identity management mechanism is required for providing effective addressing to global devices, thus making all the connected nodes "similar" in terms of protocols and encodings. Some efforts have been carried out to transform the heterogeneity of the fruitage data into homogeneity formats for arable farming such as ISO 11783 (ISOBUS) standards designed by Agricultural Industry Electronics Foundation (AEF) (Fountas et al., 2015). Additionally, the usage of middleware applications like SmartFarmNet or FIWARE, HYDRA, and SMEPP can lessen the data heterogeneity problem in the fruitage industry to ensure standardization (Kour and Arora, 2020). Some reference models like Industrial Internet Reference Architecture (IIRA), Reference Architecture Model Industries (RAMI 4.0), and the European Research Cluster on IoT (IERC) are working in this direction to make IoT a global technology. Also, protected agriculture requires remote monitoring and standard IP (Ingress Protection) ratings for the proper functioning of the sensors in arable farming (Baranwal et al., 2020). International standards (IEC 60529) define the IP ratings for the life span of the IoT sensors.
- **Cyber security Challenges**: With increasing network devices, the probability of digital attacks and vulnerability increases (Shafiq et al., 2020d). The real-time sensors and wireless network employed for fruitage disease classification are deployed in unsupervised and open farmland scenarios. Technically, the following open challenges need to be addressed: (1) trust mechanism; (2) communication and user data privacy and security; (3) services and application security. Such IoT management is susceptible to interference from outside sources and unauthorized users. Thus, multiple data privacy mechanisms such as encryption & decryption, identity authorization mechanism, and anonymity mechanism are required for protected agriculture (Shafiq et al., 2020a). Agriculture 4.0 has adopted cloud-based smart applications, opening up new potential for adversaries to oscillate cyberattacks. Typically, conventional end-to-end encryption and decryption techniques are used to secure the data. However, IoT devices require energy-efficient and least computational power encryption algorithms as the sensors constantly collect real-time fruitage data for a long time using battery power. Also, data integrity, authorization, and authentication mechanisms are the major concerns for secure communication between devices. The literature survey concluded that the real-time fruitage information has been stored on the PostgreSQL database and Neo4j database, which are vulnerable to SQL injection attacks (Zhu et al., 2023; Akhter and Sofi, 2021). On the same note, the database access management methods should be reinforced by restricting the database usage on the basis of the security level. Also, the fruitage images have been compressed and uncompressed using the DCT algorithm, hybrid measurement matrix method, and OMP method (Orthogonal matching pursuit) (Nandhini et al., 2015).
- **Resource Optimization**: As trillions of nodes are connected to generate and store massive fruitage data using real-time sensors. Thus, efficient routing strategies are required to optimize the allocation of sensor nodes to the gateway nodes at low communication costs and energy consumption. The literature survey witnessed the Sunflower EarthWorm (S-EWA) method to optimize the routing path between the sensor nodes for data propagation in the rice agriculture farmlands (Vimala et al., 2021). The optimized IoT system must ensure load balancing, traffic management, and minimum Service Level Agreement (SLA) violations. The IoT system consisting of Jetson Xavier Nx and TensorRT accelerator for edge computing has further increased the recognition and validation accuracy of the disease recognition model (Zhang et al., 2022). The Thinkspeak IoT cloud services have been addressed to store enormous fruitage data because of open source and data scalable features (Pawara et al., 2018; Microsoft, 2022). ThingSpeak also offers incorporation with IoT services, platforms, and applications using its RESTful API. Cutting-edge technologies like cloud-based services, fog-edge-based computing and federated learning algorithms can be utilized to store information in the cloud. Furthermore, the Thinkspeak cloud via SIM800 Quad-band GSM/GPRS modem with SMA (SubMiniature Version A) connector has been exploited to store the temperature and humidity data on the cloud. Despite this, there are several open challenges to be addressed; (1) data collection, (2) data decentralization, and (3) lack of decision support infrastructure. Non-destructive swarm robotics utilizes the self-organizing, decentralized algorithms that may be employed for disease detection in arable farming (Albiero et al., 2022). These techniques might be equipped with multi-sensors, including the hyperspectral, and NIR cameras for fruit & vegetable image acquisition, and effectively gather information about the disease categories of vegetables and fruits.

8.4. Fruitage dataset research gaps and future directions

- **Unavailability of real-time benchmark fruitage dataset**: Despite the availability of extremely complicated models, fine-tuning the architectures on fruitage images under in-field scenarios is necessary for their proper deployment in a real-time environment. The literature survey acknowledged that the scarcity of fruitage diversity regarding image features, shape, size, disease, and multiple species would result in poor performance of the model deployment. Significant efforts may be required to foster collaboration and consensus among multiple research teams in the collection and sharing of fruitage datasets, which can enhance both the quantity and quality of these valuable fruitage datasets. Generally, most in-field datasets emphasize a single fruit or vegetable, leading to the future direction for developing benchmark datasets for multiple cultivars with in-field scenarios. Most of the recent studies witnessed the dataset containing single fruit or vegetable cultivars known as single-fruitage dataset (Sara et al., 2022; Kaufmane et al., 2022; Alencastre-Miranda et al., 2018). Still, a few research works also focussed on the multi-fruitage dataset to build the robust disease recognition system (Meshram and Patil, 2022; Medhi and Deb, 2022). The captured dataset size is very small, which makes the deep learning model hard to train (Rajbongshi et al., 2022; Rauf et al., 2019; Kumar et al., 2021). Furthermore, the banana and date palm images have been collected under natural light, respectively, to witness the real-field data acquisition (Medhi and Deb, 2022; Altaheri et al., 2019). Additionally, some research studies utilized non-public or internet images of infected fruits or vegetables; thus, result replication is a challenging task due to the inaccessibility of fruitage dataset (Zhu et al., 2023; Oraño et al., 2019). Thus, there is a need for a robust real-field dataset that contains mostly consumed fruits and vegetables from different cultivars with different growing stages (flowering to maturation). Also, it is recorded that computer vision-based video datasets are rarely available to train the models. In the future, developing video fruitage datasets may be a demanding asset for the researcher's community working in the area. Moreover, the fruitage dataset developed from the satellite images could play a vital role in predicting the disease maps under different environmental conditions.

- **Diverse and complex background**: Real-time fruitage disease detection is a challenging task due to environmental factors like cloudy weather, lightning conditions, and changeable backdrops. Moreover, identifying diseases with a moving camera, such as UAV or satellite imaging, is more difficult than static images due to shifting background conditions. Thus, robust foreground and background separation and segmentation techniques must be explored for better results. Also, most of the recent research has been concentrated on datasets collected under controllable laboratory circumstances, which becomes the foundation for future work. The fruitage images should be gathered in the real-field arable farmlands to enhance the generalization ability of the robust disease classification model. The survey witnessed that the fruitage image acquisition has been carried out using digital cameras, Raspberry Pi cameras, and thermal cameras under a controlled environment to enhance the disease recognition accuracy (Tariq et al., 2022; Meshram and Patil, 2022; Alencastre-Miranda et al., 2018). Thus, the resultant dataset contains processed images without complex backgrounds, fixed backgrounds, or white backgrounds. Thus, intelligent models find it hard to predict disease in fruits or vegetables during real-time scenarios. To overcome this challenge, novel methods may be developed for automatic background noise removal from the multiple fruit images (Oltean, 2021). Graph-based segmentation methods can be employed that utilize graph cut techniques to segregate the foreground images from the complex background. GAN-based dehazing methods may be further explored to mitigate the disease classification problem under extreme weather conditions.
- **Close-distance Fruitage images**: The majority of the publicly accessible datasets contain mobile camera-captured images taken from a close distance. Also, disease-prone areas comprise a sizeable portion of an image, irrespective of size. However, for the real-time performance of the models, smart, innovative models are required, which are trained on datasets containing proximity images from the camera. Moreover, the datasets captured from mobile devices do not cover the image information from different angles, which may leave important feature information for disease classification. The literature survey has emphasized the in-situ fruitage dataset collection with an appropriate angle to exaggerate the detailed spatial information (Rajbongshi et al., 2022; Kumar et al., 2021). It is recommended to utilize digital twin, binocular stereo vision, and 3D depth sensors to create the virtual visualization of fruits and vegetables, thus enabling the detailed disease information (Tang et al., 2023; Shoji et al., 2022). The study also concludes that the UAV-captured datasets will be the future demands for developing robust agricultural applications. Spraying Agri-robots is the future dimension to identify diseases on the fruit or vegetables and then spray targeted pesticide treatment to affected species. Furthermore, these robots can be employed for other tasks, such as crop harvesting and lifting. They can be controlled using remote sensors via smartphones for continuous fruit disease monitoring.
- **Sample size symptoms (SSS) recognition**: Diagnosing the symptoms early is still a challenge for the researchers due to minute symptoms. Tiny lesions in fruits & vegetables can cumulatively degrade their quality and shelf life, thus raising concerns about food safety. Therefore, models should be robust enough to catch the tiniest symptoms for effective classification. Deep reinforcement learning methods like Trust Region Policy Optimization (TRPO), Proximal Policy Optimization (PPO), and Deep Q-Networks (DQN) may be deployed to identify the tiny infected regions present on the outer surface of the fruits and vegetables. Similarly, non-destructive techniques such as Computed Tomography (CT), and NIR spectroscopy may be employed for identifying the internal lesions in fruits & vegetables. Gas sensors are an indirect IoT approach in identifying internal fruit defects by finding the variation in gases emitted by fruits or vegetables as they ripen or develop defects (Alam et al., 2021). These can be integrated with spectroscopy techniques to develop Gas Chromatography and Mass Spectrometry (GC–MS) to determine the internal tiny defects in the fruits or vegetables. A few IoT instances, such as RFID tags, ultrasonic transducers, piezoelectric sensors, and acoustic sensors, may be deployed to further explore the attributes of the fruit in real-field farmlands (Mukhopadhyay et al., 2021).
- **Image Resolution Diversity**: Fruitage image diversity plays a crucial role in enhancing both the spatial and spectral resolutions of a dataset. The literature survey witnessed that the RGB cameras are the most commonly used sensors, capable of capturing high spatial resolution images in the visible spectrum (Altaheri et al., 2019; Oltean, 2021; Gené-Mola et al., 2021). They excel in providing rich details related to color, and texture features, thus invaluable for multiple disease classification. On the contrary side, RGB images are highly susceptible to lighting conditions and can only capture information within the visible spectrum. Recently, hyperspectral, multispectral and thermal imagery have gained prominence attention in fruitage sensing due to their ability to reveal information in the invisible spectrum, which proves invaluable for early disease detection in the fruit or vegetables (Maimaitijiang et al., 2020; Steinbrener et al., 2019). Additionally, satellite-based Synthetic Aperture Radar (SAR) and optical images have shown notable presence in large-scale farmland mapping. Thus, future datasets should be curated by combining data from multiple camera sensors. This approach helps to offset the resolution limitations inherent in individual sensor types, ensuring a more comprehensive and informative dataset for disease identification applications. Spectral Disease Indices (SDI), Plant Disease Index (PDI), and Vegetation Indices (VI) are the important parameters used to increase the classification accuracy of disease identification and differentiation under real-world scenarios (Al-Saddik et al., 2017).

9. Summary and discussion

This paper summarizes the current contributions and state-of-the-art development in machine learning, deep learning, and the Internet-of-Things, for fruit and vegetable disease identification, sorting, and grading. Fruit and vegetable safety, hygiene, and nutritional value are critical considerations in reducing food waste. The excellent quality of the fruitage must be monitored, and it must also be protected from deterioration and decay caused by climate variables such as humidity, temperature, and illumination. In this study, comparable fruit and vegetable disease classification tools using smart techniques have been discussed. It has covered ecological aspects such as temperature, moisture, alcohol web content, and light exposure for fruits and vegetables. Most of the smart technologies coupled with visualization methods achieve a classification accuracy of more than 90% in the disease classification domain for multiple cultivars of fruits and vegetables. Fig. 5 depicts the accuracy achieved using different intelligent models for the disease identification of fruits and vegetables. It is evident from the figure that fused technology has outperformed conventional methods to attain maximum accuracy. Concisely, SVM and clustering algorithms in the machine learning domain, pre-trained CNN using DTL and their variants in the deep learning domain, and UAV and microcontrollers (Arduino and Raspberry PI) in the IoT become the hotspot for fruit and vegetable disease detection and classification.

Moreover, the consolidated report of the studied models has also been presented in Fig. 6 regarding classification accuracy, precision, sensitivity, and specificity as the performance attributes. The Y-axis represents the contribution percentage (the number of studies in which the fruit is covered divided by the total number of studies) of a particular species, while the X-axis denotes the percentage of studies

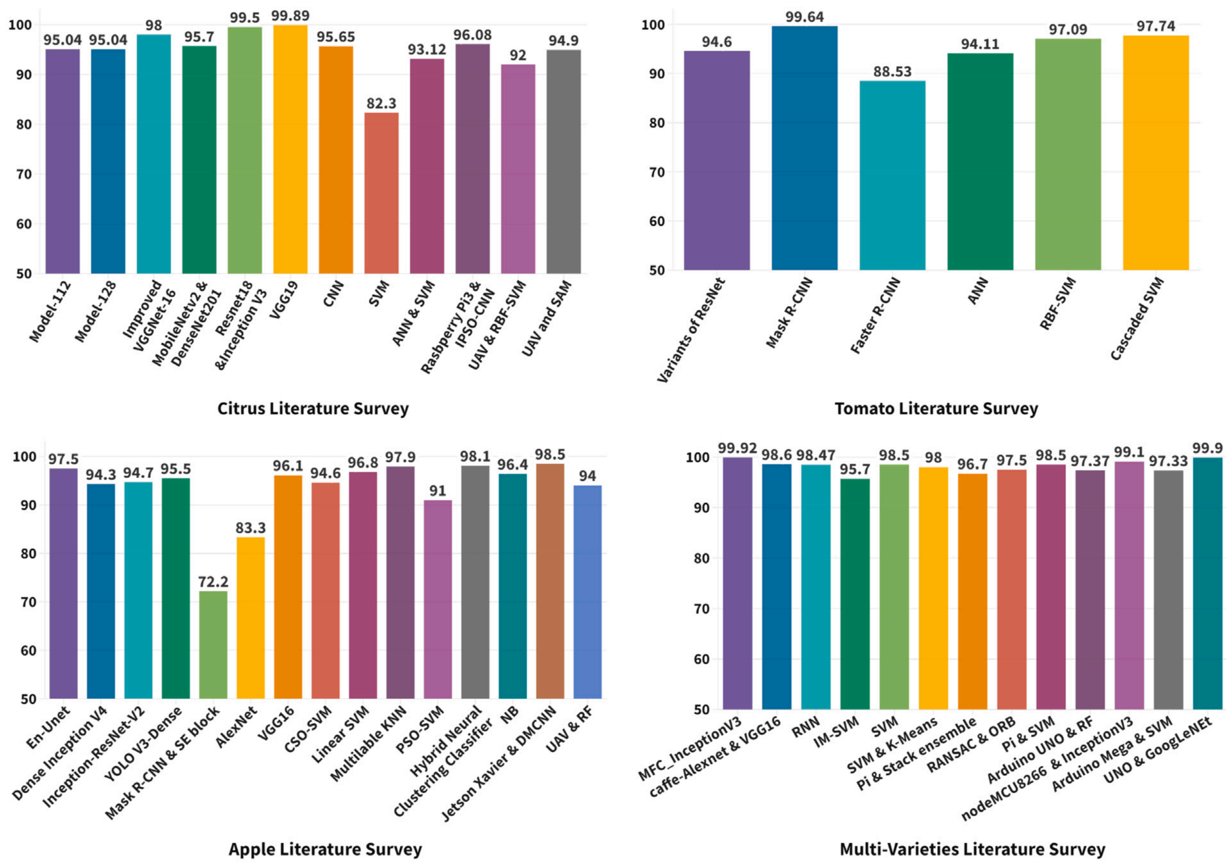

Fig. 5. Accuracy comparison chart for primary fruits.

in which a particular parameter has been considered for evaluation. For instance, as depicted in the figure, apple tissues have been used in 25.2% of the total considered studies. Whereas, 72.7% apple papers calculated the accuracy as the performance parameter. As shown in the figure, the apple is the most researched topic for disease classification in agriculture. At the same time, researchers have shown the least interest in fruits and vegetables like jackfruit, guava, potato, jute, pear, and strawberry in the miscellaneous section. It is clear from the figure that accuracy and Kappa's coefficient are considered the most and least important evaluation criteria for disease classification, respectively. Some of the other performance measures for fruit and vegetable disease detection are False Positive Rate (FPR), False Negative Rate (FNR), Matthews Correlation Coefficient (MCC), ROC curve, number of parameters, silhouette score, and segmentation accuracy.

It is investigated from the study that several methods have used hand-crafted feature extraction approaches for extracting shape, color, size, and texture information from an image (Shafiq et al., 2020c). Recently, deep learning-based models have received significant attention from researchers for feature extraction due to their ability to extract features automatically without human interference, giving crucial information. It is important to note that the number of layers in the deep models varies depending on the dataset's dimensions. For the high-dimensional dataset, a large number of layers and filters are required to extract the important features. During the image acquisition phase, several studies have collected high-resolution images using UAV, near-ground techniques, hyperspectral imaging, near-infrared imaging, multispectral imaging, RGB cameras, and other imaging techniques to obtain high disease classification accuracy. However, satellite technologies are restricted in their spatial/temporal resolutions and limited viewing capabilities in harsh weather conditions. Similarly, near-ground techniques have limited area coverage and need considerable effort to cover broad landscapes. Therefore, UAVs paired with sophisticated sensing technologies might be a cost-effective method for detecting fruit and vegetable diseases in small, medium, and large agricultural areas, enabling farmers to apply the correct treatment at the proper place, in the appropriate quantity, and at the right time.

Furthermore, the privacy of real-field data is one of the primary concerns for agrarians. A huge amount of data is required to train and test the models. Sometimes, farmers do not want to share the data due to privacy reasons, or the data source is heterogeneous. To overcome such issues, the federated learning concept has been introduced to ensure data privacy, partitioning, and system heterogeneity. Moreover, deep learning models require high computation resources and storage devices for effective disease prediction. Also, the majority of the farmers from developing nations do not have the advanced skills to become technology-friendly. Thus, providing cost-effective and user-friendly solutions to farmers is the future step for researchers in academia and industries. Li and Chao (2020) suggested that the regular convolution network performs poorly on the real-field data due to the forgetting issue, which can be mitigated using continual or incremental learning for effective disease classification. Moreover, regular neural network methods act as black-box machines focusing on the input and output without knowing the output interpretation. XAI algorithms may be integrated with deep learning models to enhance the comprehension and transparency of the disease classification models (Bhandari et al., 2023). Also, it is noted that the autonomous disease classification system requires a hardware and software prototype. The hardware module can be structured using IoT devices for real-time image acquisition, such as humidity, pressure, temperature, and nutrition index in fruits and vegetables. Similarly, the software framework can be integrated with deep learning algorithms for feature extraction and disease prediction.

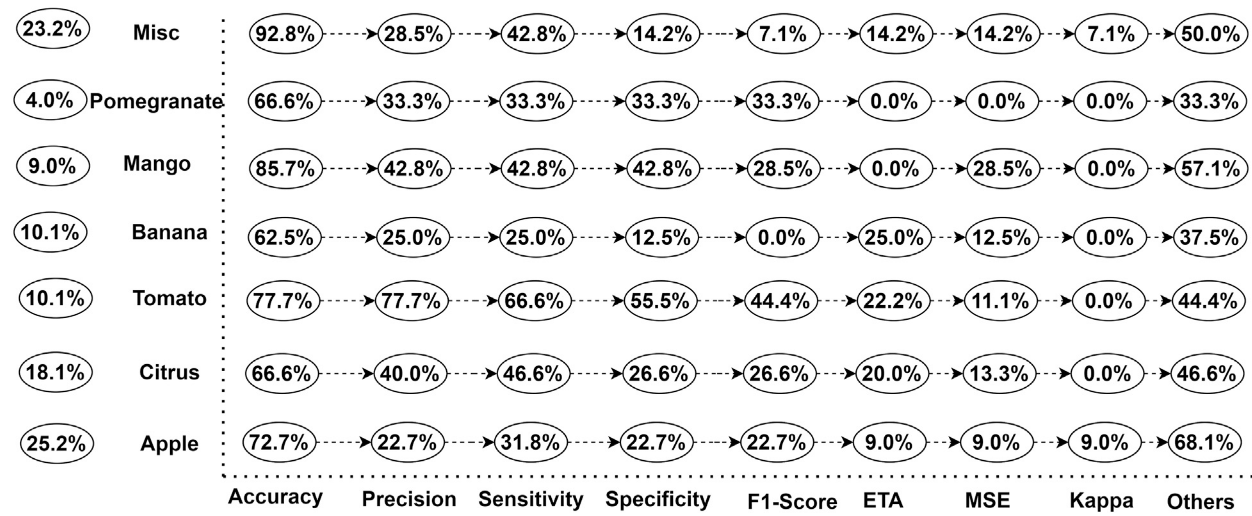

Fig. 6. Analysis of different performance metrics for fruits and vegetables covered in SLR.

10. Conclusion

The primary objective of "Agri-Food 4.0" is not limited to growing fruitage crops. It gives way to a new "smart" context in the vegetative phase by introducing ICT technologies to monitor the crops and predict the loss. As a result, farms can shift away from the conventional supply-oriented, expensive, unidentified approach toward a valuation, information-driven process in which availability and demand are constantly matched. As a result, farms and food supply chains could develop into self-adaptive ecosystems where intelligent, self-reliant objects could operate, make decisions, and even adapt without human interaction. This study highlights the recent advancements in machine learning, deep learning, and IoT for fruitage disease recognition and classification, detecting internal and external lesions to fruits, and their level of maturity. Ninety-nine peer-reviewed articles from the last six years are covered in the study to unfold major research questions, challenges, and opportunities. The study presents a detailed assessment of work on mutually acceptable frameworks through a range of research objectives. Moreover, the study covers the technology-wise research, challenges, opportunities, gaps, and future directions for fruitage disease detection and monitoring. Moreover, the publicly available fruit and vegetable datasets, their limitations, and future opportunities are also detailed. The study finds that several high-performance design models are tested for early disease detection on publicly available datasets, primarily taken in controlled laboratory environmental scenarios. However, the in-field implementation of such systems is still limited due to high computational costs and scarcity of resources, especially in developing countries. It is also recorded that technical attributes (resolution, illumination), dataset collection scenarios (laboratory, in-field), and feature attributes (texture, energy, size) have a significant impact on spectral reflectance. Additional reflectance analysis based on crop vegetation indices is an additional future space during crop growth at all stages of the disease. It is noted that the fusion of different sensible techniques leads to an efficient disease classification framework, which may vary by author perception, working environment, and datasets. Generally, fusion methods involve image acquisition from multi-sensors, satellites, camera modules, UAVs, robotics, image pre-processing using computer vision techniques, and disease classification and prediction using DL and ML models. Besides, it is tough to deduce the superiority of one approach over the other. It is noted that every piece of cropland is crucial to maximize the yield and minimize the loss. This study may serve as a tool for the researchers working in the area to get consolidated in place information about recent advancements, limitations, and future opportunities in the field of the agri-fruit industry.

Declaration of competing interest

The authors certify that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Aasha Nandhini, S., Hemalatha, R., Radha, S., Indumathi, K., 2018. Web enabled plant disease detection system for agricultural applications using WMSN. Wirel. Pers. Commun..

Abdulridha, J., Batuman, O., Ampatzidis, Y., 2019. UAV-based remote sensing technique to detect citrus canker disease utilizing hyperspectral imaging and machine learning. Remote Sens..

Agriculture, S., 2022. Smart agriculture market report share, size, penetration forecast 2028. https://www.zionmarketresearch.com/report/smart-agriculture-market, (Accessed on 08/12/2022).

Akhter, R., Sofi, S.A., 2021. Precision agriculture using IoT data analytics and machine learning. J. King Saud Univ. Comput. Inf. Sci..

Al-Saddik, H., Simon, J.-C., Cointault, F., 2017. Development of spectral disease indices for 'flavescence dorée' grapevine disease identification. Sensors.

Al-Sanabani, D.G.A., Solihin, M.I., Pui, L.P., Astuti, W., Ang, C.K., Hong, L.W., 2019. Development of non-destructive mango assessment using handheld spectroscopy and machine learning regression. In: Journal of Physics: Conference Series. IOP Publishing.

Alahmed, A., Alrasheedi, A., Alharbi, M., Alrebdi, N., Aleasa, M., Moulahi, T., 2020. Machine learning for real-time data reduction in cloud of things. In: 2020 2nd International Conference on Computer and Information Sciences. ICCIS, IEEE.

Alam, A.U., Rathi, P., Beshai, H., Sarabha, G.K., Deen, M.J., 2021. Fruit quality monitoring with smart packaging. Sensors.

Alam Siddiquee, K.N.E., Islam, M.S., Dowla, M.Y.U., Rezaul, K.M., Grout, V., 2020. Detection, quantification and classification of ripened tomatoes: a comparative analysis of image processing and machine learning. IET Image Process..

Albahri, A., Duhaim, A.M., Fadhel, M.A., Alnoor, A., Baqer, N.S., Alzubaidi, L., Albahri, O., Alamoodi, A., Bai, J., Salhi, A., et al., 2023. A systematic review of trustworthy and explainable artificial intelligence in healthcare: Assessment of quality, bias risk, and data fusion. Inf. Fusion.

Albiero, D., Garcia, A.P., Umezu, C.K., de Paulo, R.L., 2022. Swarm robots in mechanized agricultural operations: A review about challenges for research. Comput. Electron. Agric..

Alencastre-Miranda, M., Davidson, J.R., Johnson, R.M., Waguespack, H., Krebs, H.I., 2018. Robotics for sugarcane cultivation: Analysis of billet quality using computer vision. IEEE Robot. Autom. Lett..

Almadhor, A., Rauf, H.T., Lali, M.I.U., Damaševičius, R., Alouffi, B., Alharbi, A., 2021. AI-driven framework for recognition of guava plant diseases through machine learning from DSLR camera sensor based high resolution imagery. Sensors.

Altaheri, H., Alsulaiman, M., Faisal, M., Muhammed, G., 2019. Date fruit dataset for automated harvesting and visual yield estimation. In: Proc. IEEE DataPort.

Alzubaidi, L., Bai, J., Al-Sabaawi, A., Santamaría, J., Albahri, A., Al-dabbagh, B.S.N., Fadhel, M.A., Manoufali, M., Zhang, J., Al-Timemy, A.H., et al., 2023. A survey on deep learning tools dealing with data scarcity: definitions, challenges, solutions, tips, and applications. J. Big Data.
Antony, M.A., Kumar, R.S., 2021. A comparative study on predicting food quality using machine learning techniques. In: 2021 7th International Conference on Advanced Computing and Communication Systems. ICACCS, IEEE.
APEDA, 2022. Fresh fruits and vegetables. https://apeda.gov.in/apedawebsite/six_head_product/FFV.htm, (Accessed on 10/19/2022).
Arya, P.S., Gangwar, M., 2021. A proposed architecture: Detecting freshness of vegetables using internet of things (IoT) & deep learning prediction algorithm. In: 2021 3rd International Conference on Advances in Computing, Communication Control and Networking. ICAC3N, IEEE.
ATmega, 2022. Atmega2560 | microchip technology. https://www.microchip.com/en-us/product/ATmega2560, (Accessed on 08/19/2022).
Baranwal, G., Singh, M., Vidyarthi, D.P., 2020. A framework for IoT service selection. J. Supercomput..
Behera, S.K., Rath, A.K., Mahapatra, A., Sethy, P.K., 2020. Identification, classification & grading of fruits using machine learning & computer intelligence: a review. J. Ambient Intell. Humaniz. Comput..
Bellet, L.R., 2006. Ergodic properties of Markov processes. In: Open Quantum Systems II. Springer.
Bhandari, M., Shahi, T.B., Neupane, A., Walsh, K.B., 2023. Botanicx-AI: Identification of tomato leaf diseases using an explanation-driven deep-learning model. J. Imaging.
Bhargava, A., Bansal, A., 2020a. Automatic detection and grading of multiple fruits by machine learning. Food Anal. Methods.
Bhargava, A., Bansal, A., 2020b. Machine learning based quality evaluation of mono-colored apples. Multimedia Tools Appl..
Bhargava, A., Bansal, A., 2020c. Quality evaluation of mono & bi-colored apples with computer vision and multispectral imaging. Multimedia Tools Appl..
Bhole, V., Kumar, A., Bhatnagar, D., 2020. A texture-based analysis and classification of fruits using digital and thermal images. In: ICT Analysis and Applications. Springer.
Bishop, C.M., Nasrabadi, N.M., 2006. Pattern Recognition and Machine Learning, vol. 4, Springer, pp. 4–23.
Buyukarikan, B., Ulker, E., 2022. Classification of physiological disorders in apples fruit using a hybrid model based on convolutional neural network and machine learning methods. Neural Comput. Appl..
Caleb, O.J., Mahajan, P.V., Al-Said, F.A.-J., Opara, U.L., 2013. Modified atmosphere packaging technology of fresh and fresh-cut produce and the microbial consequences—a review. Food Bioprocess Technol..
Chandrasekaran, I., Panigrahi, S.S., Ravikanth, L., Singh, C.B., 2019. Potential of near-infrared (NIR) spectroscopy and hyperspectral imaging for quality and safety assessment of fruits: An overview. Food Anal. Methods.
Chaturvedi, A., Sharma, S., Janghel, R.R., 2023. Detection of external defects in tomatoes using deep learning. J. Ambient Intell. Humaniz. Comput..
Chaudhary, A., Kolhe, S., Kamal, R., 2016. An improved random forest classifier for multi-class classification. Inf. Process. Agric..
Chen, H., Lundberg, S.M., Lee, S.-I., 2022. Explaining a series of models by propagating Shapley values. Nat. Commun..
Cheng, Y., 1995. Mean shift, mode seeking, and clustering. IEEE Trans. Pattern Anal. Mach. Intell..
CISCO, 2022. Cisco annual internet report - cisco. https://www.cisco.com/c/en/us/solutions/executive-perspectives/annual-internet-report/index.html, (Accessed on 08/12/2022).
Cortes, C., Vapnik, V., 1995. Support-vector networks. Mach. Learn..
da Costa, A.Z., Figueroa, H.E., Fracarolli, J.A., 2020. Computer vision based detection of external defects on tomatoes using deep learning. Biosyst. Eng..
Cugmas, B., Štruc, E., 2020. Accuracy of an affordable smartphone-based teledermoscopy system for color measurements in canine skin. Sensors.
da Silveira, F., Lermen, F.H., Amaral, F.G., 2021. An overview of agriculture 4.0 development: Systematic review of descriptions, technologies, barriers, advantages, and disadvantages. Comput. Electron. Agric..
Dai, Q.-y., Zhang, C.-p., Wu, H., 2016. Research of decision tree classification algorithm in data mining. Int. J. Database Theory Appl..
Dairath, M.H., Akram, M.W., Mehmood, M.A., Sarwar, H.U., Akram, M.Z., Omar, M.M., Faheem, M., 2023. Computer vision-based prototype robotic picking cum grading system for fruits. Smart Agric. Technol..
Das, S., Kar, S., Sekh, A.A., 2020. Fgrade: A large volume dataset for grading tomato freshness quality. In: International Conference on Computer Vision and Image Processing. Springer.
De'ath, G., Fabricius, K.E., 2000. Classification and regression trees: a powerful yet simple technique for ecological data analysis. Ecology.
Devi, D., Anand, A., Sophia, S., Karpagam, M., Maheswari, S., 2020. IoT-deep learning based prediction of amount of pesticides and diseases in fruits. In: 2020 International Conference on Smart Electronics and Communication. ICOSEC, IEEE.
Dhaka, V.S., Meena, S.V., Rani, G., Sinwar, D., Ijaz, M.F., Woźniak, M., 2021. A survey of deep convolutional neural networks applied for prediction of plant leaf diseases. Sensors.
Dhiman, P., Kaur, A., Hamid, Y., Alabdulkreem, E., Elmannai, H., Ababneh, N., 2023. Smart disease detection system for citrus fruits using deep learning with edge computing. Sustainability.
Dhiman, P., Kukreja, V., Manoharan, P., Kaur, A., Kamruzzaman, M., Dhaou, I.B., Iwendi, C., 2022a. A novel deep learning model for detection of severity level of the disease in citrus fruits. Electronics.
Dhiman, B., Kumar, Y., Hu, Y.-C., 2021. A general purpose multi-fruit system for assessing the quality of fruits with the application of recurrent neural network. Soft Comput..
Dhiman, B., Kumar, Y., Kumar, M., 2022b. Fruit quality evaluation using machine learning techniques: review, motivation and future perspectives. Multimedia Tools Appl..
Doh, B., Zhang, D., Shen, Y., Hussain, F., Doh, R.F., Ayepah, K., 2019. Automatic citrus fruit disease detection by phenotyping using machine learning. In: 2019 25th International Conference on Automation and Computing. ICAC, IEEE.
Drugan, M.M., 2019. Reinforcement learning versus evolutionary computation: A survey on hybrid algorithms. Swarm Evol. Comput..
Dubey, A.K., Ratan, R., Rocha, A., et al., 2020. Computer vision based analysis and detection of defects in fruits causes due to nutrients deficiency. Cluster Comput..
Dubey, A.K., Rocha, A., 2023. A non-invasive approach for calcium deficiency detection in pears using machine learning. Neural Comput. Appl..
Dybå, T., Dingsøyr, T., 2008. Empirical studies of agile software development: A systematic review. Inf. Softw. Technol..
El-aziz, A., Atrab, A., Darwish, A., Oliva, D., Hassanien, A.E., 2020. Machine learning for apple fruit diseases classification system. In: The International Conference on Artificial Intelligence and Computer Vision. Springer.
El-Dosuky, M.A., Oliva, D., Hassanien, A.E., 2020. An artificial intelligence system for apple fruit disease classification based on support vector machine and cockroach swarm optimization. In: The International Conference on Artificial Intelligence and Computer Vision. Springer.
FAO, 2022a. At least 40% global crops lost to pests every year: FAO. https://www.downtoearth.org.in/news/agriculture/at-least-40-global-crops-lost-to-pests-every-year-fao-77252, (Accessed on 09/15/2022).
FAO, 2022b. SDG 2. Zero hunger sustainable development goals food and agriculture organization of the united nations. https://www.fao.org/sustainable-development-goals/goals/goal-2/en/, (Accessed on 10/19/2022).
Fenu, G., Malloci, F.M., 2021. DiaMOS plant: A dataset for diagnosis and monitoring plant disease. Agronomy.
Floerkemeier, C., Lampe, M., 2005. RFID middleware design: addressing application requirements and rfid constraints. In: Proceedings of the 2005 Joint Conference on Smart Objects and Ambient Intelligence: Innovative Context-Aware Services: Usages and Technologies.
Fountas, S., Sorensen, C.G., Tsiropoulos, Z., Cavalaris, C., Liakos, V., Gemtos, T., 2015. Farm machinery management information system. Comput. Electron. Agric..
Garg, A., Mago, V., 2021. Role of machine learning in medical research: A survey. Comp. Sci. Rev..
Gawas, A., Tolia, K., Gaikwad, S., Deshpande, K., Upadhyaya, K., Jain, S., 2021. E-fresh: Computer vision and IOT framework for fruit freshness detection. In: 2021 International Conference on Advances in Computing, Communication, and Control. ICAC3, IEEE.
GDP, I., 2022. India GDP sector-wise StatisticsTimes.com. https://statisticstimes.com/economy/country/india-gdp-sectorwise.php, (Accessed on 10/19/2022).
Gené-Mola, J., Sanz-Cortiella, R., Rosell-Polo, J.R., Escolà, A., Gregorio, E., 2021. Pfuji-size dataset: A collection of images and photogrammetry-derived 3D point clouds with ground truth annotations for fuji apple detection and size estimation in field conditions. Data Brief.
GHI, 2022. India - global hunger index (GHI) - peer-reviewed annual publication designed to comprehensively measure and track hunger at the global, regional, and country levels. https://www.globalhungerindex.org/india.html, (Accessed on 10/17/2022).
Ghodke, V., Pungaiah, S.S., Shamout, M.D., Sundarraj, A.A., Judder, M.I., Vijayprasath, S., 2022. Machine learning for auto segregation of fruits classification based logistic support vector regression. In: 2022 2nd International Conference on Technological Advancements in Computational Sciences. ICTACS, IEEE.
Goldstein, B.A., Navar, A.M., Carter, R.E., 2017. Moving beyond regression techniques in cardiovascular risk prediction: applying machine learning to address analytic challenges. Eur. Heart J..
Gollakota, A., Srinivas, M., 2011. Agribot—A multipurpose agricultural robot. In: 2011 Annual IEEE India Conference. IEEE.
Habib, M., Mia, M., Uddin, M.S., Ahmed, F., et al., 2020. A computer vision approach for jackfruit disease recognition. In: Proceedings of International Joint Conference on Computational Intelligence. Springer.
Hamilton, N., Webb, A., Wilder, M., Hendrickson, B., Blanck, M., Nelson, E., Roemer, W., Havens, T.C., 2022. Enhancing visualization and explainability of computer vision models with local interpretable model-agnostic explanations (LIME). In: 2022 IEEE Symposium Series on Computational Intelligence. SSCI, IEEE.
Han, C.H., Kim, E., Doan, T.N.N., Han, D., Yoo, S.J., Kwak, J.T., 2021. Region-aggregated attention CNN for disease detection in fruit images. PLoS One.
Hari, P., Singh, M.P., 2023. A lightweight convolutional neural network for disease detection of fruit leaves. Neural Comput. Appl..
Hasan, R.I., Yusuf, S.M., Mohd Rahim, M.S., Alzubaidi, L., 2023. Automatic clustering and classification of coffee leaf diseases based on an extended kernel density estimation approach. Plants.

Hasan, R.I., Yusuf, S.M., Rahim, M.S.M., Alzubaidi, L., 2022. Automated masks generation for coffee and apple leaf infected with single or multiple diseases-based color analysis approaches. Inform. Med. Unlocked.

He, K., Zhang, X., Ren, S., Sun, J., 2015. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In: Proceedings of the IEEE International Conference on Computer Vision.

Hemamalini, V., Rajarajeswari, S., Nachiyappan, S., Sambath, M., Devi, T., Singh, B.K., Raghuvanshi, A., 2022. Food quality inspection and grading using efficient image segmentation and machine learning-based system. J. Food Qual..

Hien, L.T.K., Gillis, N., 2021. Algorithms for nonnegative matrix factorization with the Kullback–Leibler divergence. J. Sci. Comput..

Honkavaara, E., Saari, H., Kaivosoja, J., Pölönen, I., Hakala, T., Litkey, P., Mäkynen, J., Pesonen, L., 2013. Processing and assessment of spectrometric, stereoscopic imagery collected using a lightweight UAV spectral camera for precision agriculture. Remote Sens..

Hotelling, H., 1933. Analysis of a complex of statistical variables into principal components. J. Educ. Psychol..

HSI, 2022. Hyperspectral imaging | high-performance devices - senop. https://senop.fi/industry-research/hyperspectral-imaging/, (Accessed on 08/20/2022).

Hu, W., Xiong, J., Liang, J., Xie, Z., Liu, Z., Huang, Q., Yang, Z., 2023. A method of citrus epidermis defects detection based on an improved YOLOv5. Biosyst. Eng..

Huawei, 2022. 2020 Global connectivity index. https://www.huawei.com/minisite/gci/en/index.html, (Accessed on 08/12/2022).

IBEF, 2023. Agriculture in India: Industry overview, market size, role in development.| IBEF. https://www.ibef.org/industry/agriculture-india, (Accessed on 01/30/2023).

ICAR, 2022. IH-sep-oct 2021.pdf. https://icar.org.in/sites/default/files/IH-Sep-Oct, (Accessed on 10/19/2022).

IFW, C., 2022. The casc ifw database. https://engineering.purdue.edu/RVL/Database/IFW/index.html, (Accessed on 09/02/2022).

ImageNet, 2023. ImageNet challenges, LSVRC. https://image-net.org/challenges/LSVRC/2013/index.php, (Accessed on 01/06/2023).

Indumathi, K., Hemalatha, R., Nandhini, S.A., Radha, S., 2017. Intelligent plant disease detection system using wireless multimedia sensor networks. In: 2017 International Conference on Wireless Communications, Signal Processing and Networking. WiSPNET, IEEE.

Iqbal, Z., Khan, M.A., Sharif, M., Shah, J.H., ur Rehman, M.H., Javed, K., 2018. An automated detection and classification of citrus plant diseases using image processing techniques: A review. Comput. Electron. Agric..

Ireri, D., Belal, E., Okinda, C., Makange, N., Ji, C., 2019. A computer vision system for defect discrimination and grading in tomatoes using machine learning and image processing. Artif. Intell. Agric..

Ismail, N., Malik, O.A., 2022. Real-time visual inspection system for grading fruits using computer vision and deep learning techniques. Inf. Process. Agric..

Jahanbakhshi, A., Kheiralipour, K., 2020. Evaluation of image processing technique and discriminant analysis methods in postharvest processing of carrot fruit. Food Sci. Nutr..

James, G.M., Sujatha, S., 2021. Categorising apple fruit diseases employing hybrid neural clustering classifier. Mater. Today: Proc..

Janarthan, S., Thuseethan, S., Rajasegarar, S., Lyu, Q., Zheng, Y., Yearwood, J., 2020. Deep metric learning based citrus disease classification with sparse data. IEEE Access.

Jawade, P., Chaugule, D., Patil, D., Shinde, H., 2020. Disease prediction of mango crop using machine learning and IoT. In: Advances in Decision Sciences, Image Processing, Security and Computer Vision. Springer.

Ji, Y., Zhao, Q., Bi, S., Shen, T., 2018. Apple grading method based on features of color and defect. In: 2018 37th Chinese Control Conference. CCC, IEEE.

Jin, X.-B., Yu, X.-H., Wang, X.-Y., Bai, Y.-T., Su, T.-L., Kong, J.-L., 2020. Deep learning predictor for sustainable precision agriculture based on internet of things system. Sustainability.

Kaggle, 2022. Fruits fresh and rotten for classification | kaggle. https://www.kaggle.com/datasets/sriramr/fruits-fresh-and-rotten-for-classification, (Accessed on 07/22/2022).

Kalia, P., Garg, A., Kumar, A., et al., 2019. Fruit quality evaluation using machine learning: A review. In: 2019 2nd International Conference on Intelligent Computing, Instrumentation and Control Technologies. ICICICT, IEEE.

Kantale, P., Thakare, S., 2020. A review on pomegranate disease classification using machine learning and image segmentation techniques. In: 2020 4th International Conference on Intelligent Computing and Control Systems. ICICCS, IEEE.

Kaufmane, E., Sudars, K., Namatēvs, I., Kalniņa, I., Judvaitis, J., Balašs, R., Strautiņa, S., 2022. QuinceSet: Dataset of annotated Japanese quince images for object detection. Data Brief.

Keele, S., et al., 2007. Guidelines for Performing Systematic Literature Reviews in Software Engineering. Tech. Rep..

Khalifa, N.E., Loey, M., Mirjalili, S., 2022. A comprehensive survey of recent trends in deep learning for digital images augmentation. Artif. Intell. Rev..

Khan, M.A., Akram, T., Sharif, M., Awais, M., Javed, K., Ali, H., Saba, T., 2018. CCDF: Automatic system for segmentation and recognition of fruit crops diseases based on correlation coefficient and deep CNN features. Comput. Electron. Agric..

Khattab, A., Habib, S.E., Ismail, H., Zayan, S., Fahmy, Y., Khairy, M.M., 2019. An IoT-based cognitive monitoring system for early plant disease forecast. Comput. Electron. Agric..

Khattak, A., Asghar, M.U., Batool, U., Asghar, M.Z., Ullah, H., Al-Rakhami, M., Gumaei, A., 2021. Automatic detection of citrus fruit and leaves diseases using deep neural network model. IEEE Access.

Kizilboga, A.Y., Ergüzen, A., Erdal, E., 2020. Determination of various diseases in two most consumed fruits using artificial neural networks and deep learning techniques.

Kour, V.P., Arora, S., 2020. Recent developments of the internet of things in agriculture: a survey. IEEE Access.

Krizhevsky, A., Sutskever, I., Hinton, G.E., 2012. Imagenet classification with deep convolutional neural networks. Adv. Neural Inf. Process. Syst..

Kumar, S.D., Esakkirajan, S., Bama, S., Keerthiveena, B., 2020a. A microcontroller based machine vision approach for tomato grading and sorting using SVM classifier. Microprocess. Microsyst..

Kumar, S.D., Esakkirajan, S., Vimalraj, C., Veena, B.K., 2020b. Design of disease prediction method based on whale optimization employed artificial neural network in tomato fruits. Mater. Today: Proc..

Kumar, A., Rajpurohit, V.S., Gaikwad, N.N., 2021. Image dataset of pomegranate fruits (Punica granatum) for various machine vision applications. Data Brief.

Kumari, N., Belwal, R., et al., 2021. Hybridized approach of image segmentation in classification of fruit mango using BPNN and discriminant analyzer. Multimedia Tools Appl..

Langford, V.S., Padayachee, D., McEwan, M.J., Barringer, S.A., 2019. Comprehensive odorant analysis for on-line applications using selected ion flow tube mass spectrometry (SIFT-MS). Flavour Fragr. J..

Lee, S.H., Goëau, H., Bonnet, P., Joly, A., 2020. New perspectives on plant disease characterization based on deep learning. Comput. Electron. Agric..

LEEDS, 2022. Dataset :: Leeds butterfly dataset. http://www.josiahwang.com/dataset/leedsbutterfly/, (Accessed on 08/23/2022).

Li, D., Ahmed, F., Wu, N., Sethi, A.I., 2022. YOLO-JD: A deep learning network for jute diseases and pests detection from images. Plants.

Li, Y., Chao, X., 2020. ANN-based continual classification in agriculture. Agriculture.

Li, C., Wang, B., 2014. Fisher linear discriminant analysis. CCIS Northeastern University.

Liberati, A., Altman, D.G., Tetzlaff, J., Mulrow, C., Gøtzsche, P.C., Ioannidis, J.P., Clarke, M., Devereaux, P.J., Kleijnen, J., Moher, D., 2009. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. J. Clin. Epidemiol..

Liu, F.T., Ting, K.M., Zhou, Z.-H., 2008. Isolation forest. In: 2008 Eighth IEEE International Conference on Data Mining. IEEE.

Van der Maaten, L., Hinton, G., 2008. Visualizing data using t-SNE. J. Mach. Learn. Res..

Maimaitijiang, M., Sagan, V., Sidike, P., Hartling, S., Esposito, F., Fritschi, F.B., 2020. Soybean yield prediction from UAV using multimodal data fusion and deep learning. Remote Sens. Environ..

Mamdouh, N., Khattab, A., 2021. YOLO-based deep learning framework for olive fruit fly detection and counting. IEEE Access.

Manyika, J., Chui, M., Bisson, P., Woetzel, J., Dobbs, R., Bughin, J., D., A., 2015. The Internet of Things Mapping the Value Beyond the Hype. Mckinsey & Company.

Medhi, E., Deb, N., 2022. PSFD-musa: A dataset of banana plant, stem, fruit, leaf, and disease. Data Brief.

Memarsadeghi, N., O'Leary, D.P., 2003. Classified information: the data clustering problem. Comput. Sci. Eng..

Mendeley, 2022. Fresh and rotten fruits dataset for machine-based evaluation of fruit quality - mendeley data. https://data.mendeley.com/datasets/bdd69gyhv8/1, (Accessed on 07/22/2022).

Meshram, V., Patil, K., 2022. FruitNet: Indian fruits image dataset with quality for machine learning applications. Data Brief.

Meshram, V.A., Patil, K., Ramteke, S.D., 2021. MNet: A framework to reduce fruit image misclassification.. Ingén. Syst. Inf..

Meshram, V., Thanomliang, K., Ruangkan, S., Chumchu, P., Patil, K., 2020. Fruitsgb: top Indian fruits with quality.

Microsoft, 2022. Microsoft word - 9 IOT with cloud based smart farming for citrus fruit disease classification using optimized convolutional neu. https://researchtrend.net/ijet/pdf/IOT, (Accessed on 07/22/2022).

Mishra, S., Khatri, S.K., Johri, P., 2019. IOT based automated quality assessment for fruits and vegetables using infrared. In: 2019 4th International Conference on Information Systems and Computer Networks. ISCON, IEEE.

Mishra, P., Roger, J.M., Rutledge, D.N., Woltering, E., 2020. SPORT pre-processing can improve near-infrared quality prediction models for fresh fruits and agro-materials. Postharvest Biol. Technol..

Mohammed, M., El-Shafie, H., Munir, M., 2023. Development and validation of innovative machine learning models for predicting date palm mite infestation on fruits. Agronomy.

Mojumdar, M.U., Chakraborty, N.R., 2021. Orange & orange leaves diseases detection using computerized techniques. In: 2021 12th International Conference on Computing Communication and Networking Technologies. ICCCNT, IEEE.

Moore, D., Nauta, M.M., Evans, S.E., Rotheroe, M., 2001. Fungal Conservation: Issues and Solutions. Cambridge University Press.

Moriya, É.A.S., Imai, N.N., Tommaselli, A.M.G., Berveglieri, A., Santos, G.H., Soares, M.A., Marino, M., Reis, T.T., 2021. Detection and mapping of trees infected with citrus gummosis using UAV hyperspectral data. Comput. Electron. Agric..

Mukhopadhyay, S.C., Tyagi, S.K.S., Suryadevara, N.K., Piuri, V., Scotti, F., Zeadally, S., 2021. Artificial intelligence-based sensors for next generation IoT applications: A review. IEEE Sens. J..
Munasingha, L., Gunasinghe, H., Dhanapala, W., 2019. Identification of papaya fruit diseases using deep learning approach. In: 4th International Conference on Advances in Computing and Technology. ICACT.
Nabi, F., Jamwal, S., Padmanbh, K., 2020. Wireless sensor network in precision farming for forecasting and monitoring of apple disease: a survey. Int. J. Inf. Technol..
Nagasubramanian, G., Sakthivel, R.K., Patan, R., Sankayya, M., Daneshmand, M., Gandomi, A.H., 2021. Ensemble classification and IoT-based pattern recognition for crop disease monitoring system. IEEE Internet Things J..
Naik, B.N., Ramanathan, M., Ponnusamy, P., 2023. Refined single-stage object detection deep-learning technique for chilli leaf disease detection. J. Electron. Imaging.
Nandhini, S.A., Sankararajan, R., Rajendiran, K., 2015. Video compressed sensing framework for wireless multimedia sensor networks using a combination of multiple matrices. Comput. Electr. Eng..
Naranjo-Torres, J., Mora, M., Hernández-García, R., Barrientos, R.J., Fredes, C., Valenzuela, A., 2020. A review of convolutional neural network applied to fruit image processing. Appl. Sci..
Noguchi, S., Nishio, M., Yakami, M., Nakagomi, K., Togashi, K., 2020. Bone segmentation on whole-body CT using convolutional neural network with novel data augmentation techniques. Comput. Biol. Med..
Oltean, M., 2021. Fruits 360 dataset: new research directions.
Oppenheim, D., Shani, G., Erlich, O., Tsror, L., 2019. Using deep learning for image-based potato tuber disease detection. Phytopathology.
Oraño, J.F.V., Maravillas, E.A., Aliac, C.J.G., 2019. Jackfruit fruit damage classification using convolutional neural network. In: 2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management. HNICEM, IEEE.
Palei, S., Behera, S.K., Sethy, P.K., 2023. A systematic review of citrus disease perceptions and fruit grading using machine vision. Procedia Comput. Sci..
Park, Y., Yang, H.S., 2019. Convolutional neural network based on an extreme learning machine for image classification. Neurocomputing.
Patil, P.U., Lande, S.B., Nagalkar, V.J., Nikam, S.B., Wakchaure, G., 2021a. Grading and sorting technique of dragon fruits using machine learning algorithms. J. Agric. Food Res..
Patil, J., et al., 2021b. Pomegranate fruit diseases detection using image processing techniques: A review. Inf. Technol. Ind..
Pavel, M.I., Kamruzzaman, S.M., Hasan, S.S., Sabuj, S.R., 2019. An IoT based plant health monitoring system implementing image processing. In: 2019 IEEE 4th International Conference on Computer and Communication Systems. ICCCS, IEEE.
Pawara, S., Nawale, D., Patil, K., Mahajan, R., 2018. Early detection of pomegranate disease using machine learning and internet of things. In: 2018 3rd International Conference for Convergence in Technology. I2CT, IEEE.
Quinlan, J.R., 1996. Learning decision tree classifiers. ACM Comput. Surv..
Quy, V.K., Hau, N.V., Anh, D.V., Quy, N.M., Ban, N.T., Lanza, S., Randazzo, G., Muzirafuti, A., 2022. IoT-enabled smart agriculture: Architecture, applications, and challenges. Appl. Sci..
Rahman, S., Khan, S., Porikli, F., 2018. A unified approach for conventional zero-shot, generalized zero-shot, and few-shot learning. IEEE Trans. Image Process..
Rajbongshi, A., Sazzad, S., Shakil, R., Akter, B., Sara, U., 2022. A comprehensive guava leaves and fruits dataset for guava disease recognition. Data Brief.
Rangarajan, A.K., Whetton, R.L., Mouazen, A.M., 2022. Detection of fusarium head blight in wheat using hyperspectral data and deep learning. Expert Syst. Appl..
Rauf, H.T., Lali, M.I.U., 2021. A guava fruits and leaves dataset for detection and classification of guava diseases through machine learning. Mendeley Data.
Rauf, H.T., Saleem, B.A., Lali, M.I.U., Khan, M.A., Sharif, M., Bukhari, S.A.C., 2019. A citrus fruits and leaves dataset for detection and classification of citrus diseases through machine learning. Data Brief.
Razmjooy, N., Estrela, V.V., Loschi, H.J., 2019. A survey of potatoes image segmentation based on machine vision. In: Applications of Image Processing and Soft Computing Systems in Agriculture. IGI Global.
Refractometers, D., 2022. Digital refractometers: MA884. https://www.milwaukeeinst.com/web/products/our-products/refractometer/ma884-detail?tmpl=component&format=pdf, (Accessed on 08/29/2022).
Rehman, M.Z.U., Ahmed, F., Khan, M.A., Tariq, U., Jamal, S.S., Ahmad, J., Hussain, I., 2022. Classification of citrus plant diseases using deep transfer learning. Comput. Mater. Contin..
Reynolds, D.A., 2009. Gaussian mixture models.. Encycl. Biom..
Rizzo, M., Marcuzzo, M., Zangari, A., Gasparetto, A., Albarelli, A., 2023. Fruit ripeness classification: A survey. Artif. Intell. Agric..
Roy, K., Chaudhuri, S.S., Pramanik, S., 2021a. Deep learning based real-time industrial framework for rotten and fresh fruit detection using semantic segmentation. Microsyst. Technol..
Roy, K., Pramanik, S., Chaudhuri, S.S., 2021b. Unsupervised machine learning algorithm based rotten fruit segmentation. In: Advances in Smart Communication Technology and Information Processing. Springer.
Saha, D., Manickavasagan, A., 2021. Machine learning techniques for analysis of hyperspectral images to determine quality of food products: A review. Curr. Res. Food Sci..
Saini, A.K., Bhatnagar, R., Srivastava, D.K., 2021. Citrus fruits diseases detection and classification using transfer learning. In: Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence.
Sara, U., Rajbongshi, A., Shakil, R., Akter, B., Uddin, M.S., 2022. VegNet: An organized dataset of cauliflower disease for a sustainable agro-based automation system. Data Brief.
Saranya, S.M., Rajalaxmi, R., Prabavathi, R., Suganya, T., Mohanapriya, S., Tamilselvi, T., 2021. Deep learning techniques in tomato plant–a review. In: Journal of Physics: Conference Series. IOP Publishing.
SARSA, 2022. SARSA reinforcement learning - GeeksforGeeks. https://www.geeksforgeeks.org/sarsa-reinforcement-learning/, (Accessed on 09/06/2022).
Schoofs, H., Delalieux, S., Deckers, T., Bylemans, D., 2020. Fire blight monitoring in pear orchards by unmanned airborne vehicles (UAV) systems carrying spectral sensors. Agronomy.
Selvaraj, M.G., Vergara, A., Ruiz, H., Safari, N., Elayabalan, S., Ocimati, W., Blomme, G., 2019. AI-powered banana diseases and pest detection. Plant Methods.
Shafiq, M., Tian, Z., Bashir, A., Du, X., Guizani, M., 2020a. CorrAUC: a malicious bot-IoT traffic detection method in IoT network using machine learning techniques. IEEE Internet Things J..
Shafiq, M., Tian, Z., Bashir, A., Du, X., Guizani, M., 2020b. IoT malicious traffic identification using wrapper-based feature selection mechanisms. Comput. Secur..
Shafiq, M., Tian, Z., Bashir, A., Jolfaei, A., 2020c. Data mining and machine learning methods for sustainable smart cities traffic classification: A survey. Sustainable Cities Soc..
Shafiq, M., Tian, Z., Sun, Y., Du, X., Guizani, M., 2020d. Selection of effective machine learning algorithm and bot-IoT attacks traffic identification for internet of things in smart city. Future Gener. Comput. Syst..
Shapiro, S.C., Bona, J.P., 2010. The GLAIR cognitive architecture. Int. J. Mach. Conscious..
Sharma, A., Jain, A., Gupta, P., Chowdary, V., 2020. Machine learning applications for precision agriculture: A comprehensive review. IEEE Access.
Sharma, V., Tripathi, A.K., Mittal, H., 2022. Technological revolutions in smart farming: Current trends, challenges & future directions. Comput. Electron. Agric..
Shoji, K., Schudel, S., Onwude, D., Shrivastava, C., Defraeye, T., 2022. Mapping the postharvest life of imported fruits from packhouse to retail stores using physics-based digital twins. Resour. Conserv. Recy..
Shruthi, U., Nagaveni, V., Arvind, C., Sunil, G., 2022. Tomato plant disease classification using deep learning architectures: A review. In: Proceedings of Second International Conference on Advances in Computer Engineering and Communication Systems. Springer.
Singh, J., Kaur, H., 2018. Plant disease detection based on region-based segmentation and KNN classifier. In: International Conference on ISMAC in Computational Vision and Bio-Engineering. Springer.
Singh, A., Vaidya, G., Jagota, V., Darko, D.A., Agarwal, R.K., Debnath, S., Potrich, E., 2022. Recent advancement in postharvest loss mitigation and quality management of fruits and vegetables using machine learning frameworks. J. Food Qual..
Sivaranjani, A., Senthilrani, S., Ashok Kumar, B., Senthil Murugan, A., 2022. An overview of various computer vision-based grading system for various agricultural products. J. Horticult. Sci. Biotechnol..
Steinbrener, J., Posch, K., Leitner, R., 2019. Hyperspectral fruit and vegetable classification using convolutional neural networks. Comput. Electron. Agric..
Sugiarti, Y., Supriyatna, A., Carolina, I., Amin, R., Yani, A., et al., 2021. Model naïve Bayes classifiers for detection apple diseases. In: 2021 9th International Conference on Cyber and IT Service Management. CITSM, IEEE.
Survey, N.F.H., 2022. India.pdf. http://rchiips.org/nfhs/NFHS-5_FCTS/India.pdf, (Accessed on 10/17/2022).
Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A., 2015. Going deeper with convolutions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.
Tan, M., Le, Q., 2019. Efficientnet: Rethinking model scaling for convolutional neural networks. In: International Conference on Machine Learning. PMLR.
Tang, Y., Zhou, H., Wang, H., Zhang, Y., 2023. Fruit detection and positioning technology for a Camellia oleifera C. Abel orchard based on improved YOLOv4-tiny model and binocular stereo vision. Expert Syst. Appl..
Tariq, S., Hakim, A., Siddiqi, A.A., Owais, M., 2022. An image dataset of fruitfly species (bactrocera zonata and bactrocera dorsalis) and automated species classification through object detection. Data Brief.
Tasnim, R., Drummond, F., Zhang, Y.-J., 2021. Climate change patterns of wild blueberry fields in downeast, maine over the past 40 years. Water.
Thangaraj, R., Dinesh, D., Hariharan, S., Rajendar, S., Gokul, D., Hariskarthi, T., 2020. Automatic recognition of avocado fruit diseases using modified deep convolutional neural network. Int. J. Grid Distrib. Comput..
2022. The design of divide and conquer algorithms douglas R. Smith | download. https://ur.booksc.me/book/8550899/ddbecf, (Accessed on 08/25/2022).
Tian, Y., Li, E., Liang, Z., Tan, M., He, X., 2021. Diagnosis of typical apple diseases: A deep learning method based on multi-scale dense classification network. Front. Plant Sci..
Tian, Y., Yang, G., Wang, Z., Li, E., Liang, Z., 2019. Detection of apple lesions in orchards based on deep learning methods of cyclegan and yolov3-dense. J. Sens..
Tripathi, M.K., Maktedar, D.D., 2023. Internal quality assessment of mango fruit: an automated grading system with ensemble classifier. J. Imaging Sci..

Turkoglu, M., Hanbay, D., Sengur, A., 2019. Multi-model LSTM-based convolutional neural networks for detection of apple diseases and pests. J. Ambient Intell. Humaniz. Comput..

Ukwuoma, C.C., Zhiguang, Q., Bin Heyat, M.B., Ali, L., Almaspoor, Z., Monday, H.N., 2022. Recent advancements in fruit detection and classification using deep learning techniques. Math. Probl. Eng..

Van Engelen, J.E., Hoos, H.H., 2020. A survey on semi-supervised learning. Mach. Learn..

Vanitha, C., Malathy, S., Krishna, S., Manikantan, M., Kumar, P.D., Cibi, M.G., 2023. Detection of ripe and raw tomatoes using internet of things. In: 2023 7th International Conference on Computing Methodologies and Communication. ICCMC, IEEE.

Vasumathi, M., Kamarasan, M., 2021. An effective pomegranate fruit classification based on CNN-LSTM deep learning models. Indian J. Sci. Technol..

Vimala, S., Gladiss Merlin, N., Ramanathan, L., Cristin, R., 2021. Optimal routing and deep regression neural network for rice leaf disease prediction in IoT. Int. J. Comput. Methods.

Vincent, L., Soille, P., 1991. Watersheds in digital spaces: an efficient algorithm based on immersion simulations. IEEE Trans. Pattern Anal. Mach. Intell..

Wang, Y., Ni, X.S., 2019. Predicting class-imbalanced business risk using resampling, regularization, and model ensembling algorithms. arXiv preprint arXiv:1903.05535.

Wang, Q., Qi, F., Sun, M., Qu, J., Xue, J., 2019. Identification of tomato disease types and detection of infected areas based on deep convolutional neural networks and object detection techniques. Comput. Intell. Neurosci..

WHO, 2022. Food safety. https://www.who.int/news-room/fact-sheets/detail/food-safety, (Accessed on 10/18/2022).

Xiao, D., Pan, Y., Feng, J., Yin, J., Liu, Y., He, L., 2022. Remote sensing detection algorithm for apple fire blight based on UAV multispectral image. Comput. Electron. Agric..

Yao, N., Ni, F., Wu, M., Wang, H., Li, G., Sung, W.-K., 2022. Deep learning-based segmentation of peach diseases using convolutional neural network. Front. Plant Sci..

Yasmeen, U., Khan, M.A., Tariq, U., Khan, J.A., Yar, M.A.E., Hanif, C.A., Mey, S., Nam, Y., 2021. Citrus diseases recognition using deep improved genetic algorithm.

Yu, D., Xu, T., Song, K., 2018. Nondestructive testing of external defects on nanguo pear. Spectrosc. Lett..

Zhang, B., Gu, B., Tian, G., Zhou, J., Huang, J., Xiong, Y., 2018. Challenges and solutions of optical-based nondestructive quality inspection for robotic fruit and vegetable grading systems: A technical review. Trends Food Sci. Technol..

Zhang, T., Lei, C., Zhang, Z., Meng, X.-B., Chen, C.P., 2021. AS-NAS: Adaptive scalable neural architecture search with reinforced evolutionary algorithm for deep learning. IEEE Trans. Evol. Comput..

Zhang, W., Zhou, G., Chen, A., Hu, Y., 2022. Deep multi-scale dual-channel convolutional neural network for internet of things apple disease detection. Comput. Electron. Agric..

Zhou, X., Lee, W.S., Ampatzidis, Y., Chen, Y., Peres, N., Fraisse, C., 2021. Strawberry maturity classification from UAV and near-ground imaging using deep learning. Smart Agric. Technol..

Zhu, X.J., 2005. Semi-supervised learning literature survey.

Zhu, D., Xie, L., Chen, B., Tan, J., Deng, R., Zheng, Y., Hu, Q., Mustafa, R., Chen, W., Yi, S., et al., 2023. Knowledge graph and deep learning based pest detection and identification system for fruit quality. Internet Things.

Zhu, Y., Xie, C., Wang, G.-J., Yan, X.-G., 2017. Comparison of individual, ensemble and integrated ensemble machine learning methods to predict China's SME credit risk in supply chain finance. Neural Comput. Appl..